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BACHELOR THESIS



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Universal Deep Learning Detectors for Macromolecule Localization in Cryo-ET

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Abstract

Cryo-electron tomography (Cryo-ET) provides three-dimensional (3D) visualizations, called tomograms, of cellular environments in their native state. While these tomograms preserve vital structural context, they are often characterized by low signal-to-noise ratios, anisotropy and high annotation costs. As a result, manual particle picking does not scale, and classical template-based methods can become brittle when particle conformation, local environment or acquisition domain changes. This thesis investigates the practical value of universal deep learning (DL) detectors for macromolecule localization in cryo-ET. The work focuses on ProPicker, a promptable 3D segmentation model, and analyzes how far this family of detectors can behave as a general tool for new particle targets.

The main contribution is a reproducible experimental workflow that integrates preprocessing, prompt construction, fine-tuning, inference and evaluation. The study combines the real data cytoplasmic ribosome case from EMPIAR-10988 with a synthetic thyroglobulin dataset provided by the AIKE laboratory and generated using PolNet. The experiments show that prompt-only inference contains useful signal, but does not provide balanced performance when the domain changes. On EMPIAR-10988, fine-tuning increases the score to $F_1 = 0.5714$. On the synthetic dataset, the base model almost collapses, whereas the fine-tuned model reaches a mean F_1 of 0.891. The incremental study further shows that four training tomograms already produce a competitive regime, and that twelve tomograms with multi-prompt inference provide the best observed result, with $F_1 = 0.919$.

Finally, the thesis identifies prompt robustness as the primary remaining limitation. Even after fine-tuning, certain prompts trigger a collapse in recall, a variability that cannot be explained by signal-to-noise ratio, spatial position or simple angular rules alone. Consequently, while ProPicker does not yet qualify as a universal detector in the strictest sense, it provides a practical and adaptable foundation for the development of more general cryo-ET detection systems.

Resumen extendido (*Extended Abstract*)

Contexto y motivación.

La tomografía crioelectrónica es una técnica que permite estudiar la organización molecular en su estado nativo. A diferencia de los métodos de partícula aislada que utilizan muestras purificadas y pierden la información del entorno, la cryo-ET reconstruye volúmenes tridimensionales donde las macromoléculas se observan dentro de sus contextos biológicos originales. Esta característica es interesante en procesos celulares donde no se depende solo de la estructura de una proteína aislada, sino también de su posición, de sus interacciones locales y de la organización espacial de los complejos que la rodean. Por ello, la cryo-ET se ha convertido en una herramienta especialmente útil para conectar biología estructural y biología celular.

Sin embargo, esta capacidad conlleva retos computacionales significativos. Los tomogramas presentan una baja relación señal-ruido, anisotropía por el efecto del missing wedge y una alta densidad de estructuras donde las partículas pueden solaparse. Además las reconstrucciones suelen contener variabilidad local, diferencias de contraste, artefactos de adquisición y regiones donde la señal de la partícula queda parcialmente oculta por el entorno. Por todo eso, la localización de partículas es una fase determinante, ya que las coordenadas obtenidas definen la calidad del posterior promediado y análisis estructural. Un error en esta etapa, ya sea por omitir partículas o por incluir demasiados falsos positivos, afecta directamente a la señal estructural resultante y aumenta la carga de supervisión manual. Por tanto, el problema no consiste únicamente en detectar muchos candidatos, sino en obtener una detección suficientemente equilibrada para que las etapas posteriores puedan trabajar con datos fiables.

Problema y objetivo.

Esta tesis aborda la localización automática de macromoléculas mediante detectores basados en Deep Learning y condicionados por ejemplos o *prompts*. El objetivo es que, a partir de un volumen y un ejemplo de la partícula de interés, el sistema genere una lista de coordenadas precisas bajo una tolerancia espacial aceptable. La investigación no se limita a evaluar el rendimiento en un único conjunto de datos, sino que analiza la capacidad del modelo para mantener su utilidad al cambiar de dominio, de partícula o de condiciones de adquisición. Este concepto de detector universal se plantea como una solución práctica para reducir la dependencia de anotaciones extensas y evitar entrenamientos desde cero en cada nuevo experimento.

El término universal se utiliza aquí con cautela. No se asume que un único modelo pueda resolver todos los casos sin adaptación, sino que se estudia hasta qué punto un detector previamente entrenado puede transferir conocimiento y cuánto ajuste necesita para

funcionar en un dominio nuevo. Esta distinción es importante porque en cryo-ET los datos anotados son costosos y no siempre es viable construir grandes conjuntos de entrenamiento para cada molécula. Por ello, un detector útil debe combinar generalidad inicial, capacidad de adaptación y estabilidad ante cambios razonables del protocolo experimental.

Como modelo de referencia se ha usado ProPicker, un detector basado en segmentación 3D que permite dos modos de operación principales. El primero es la inferencia directa basada en el *prompt*, donde el modelo utiliza un ejemplo para condicionar la búsqueda de partículas similares. El segundo es el ajuste fino con un número limitado de datos, donde los pesos del modelo se adaptan al dominio objetivo. Esta estructura permite analizar la universalidad como un proceso gradual que va desde la transferencia inmediata hasta la especialización del modelo con poca supervisión. La tesis evalúa precisamente ese recorrido y estudia qué parte del rendimiento procede del modelo base, qué parte aparece tras el ajuste y qué parte depende de la calidad del ejemplo elegido como *prompt*.

Flujo experimental.

El flujo experimental se ha diseñado bajo criterios de reproducibilidad, integrando la gestión de datos, el entrenamiento y el análisis de resultados. Esta organización garantiza que factores como la conversión de coordenadas, el ajuste de contraste o la división de los conjuntos de datos no introduzcan sesgos en la comparación de resultados. En un problema volumétrico como este una diferencia pequeña en la escala de coordenadas, en el sentido del contraste o en la forma de convertir mapas de probabilidad en puntos, puede modificar de manera notable las métricas finales. Por ello, el método se construye alrededor de controles explícitos que incluyen una división fija entre entrenamiento y validación, el uso de tamaños comunes de *prompt* de $37 \times 37 \times 37$ vóxeles y una evaluación con reglas geométricas consistentes en todos los experimentos.

El repositorio se organiza para que cada experimento tenga sus datos de entrada, sus scripts, sus salidas y sus figuras. Esta separación facilita comprobar qué se ha ejecutado, qué modelo ha producido cada resultado y qué parámetros se han utilizado para calcular las métricas. También permite conservar los mapas de localización y las coordenadas intermedias, necesarios para interpretar fallos concretos. El objetivo de esta estructura no es solo generar resultados finales, sino poder reconstruir el camino que lleva a ellos.

Datos utilizados.

El estudio se apoya en dos fuentes de datos diferenciadas. Por un lado EMPIAR-10988 aporta datos reales de ribosomas citoplasmáticos para validar la infraestructura técnica en un entorno biológico representativo. Este conjunto se utiliza como prueba inicial porque permite comprobar que el flujo completo de preparación, inferencia, ajuste fino y evaluación puede ejecutarse sobre tomogramas reales. El experimento trabaja con los tomogramas *TS_029* y *TS_030* en un recorte de 128^3 vóxeles. Esta decisión reduce el coste computacional y permite depurar el procedimiento sin convertir el primer experimento en una evaluación completa del dataset EMPIAR.

Por otro lado, el dataset sintético centrado en la tiroglobulina, permite trabajar con un ground truth completo para analizar la sensibilidad a los *prompts* y la influencia de la simetría de la partícula bajo los efectos de la anisotropía. Este dataset contiene 25 tomogramas con aproximadamente 3327 partículas de la clase objetivo. Desde el segundo experimento se fija una división de 20 tomogramas para entrenamiento y 5 para validación. El uso de datos sintéticos facilita además el control sobre el reparto de entrenamiento y validación permitiendo, un análisis más sistemático de la eficiencia del modelo.

La combinación de ambos tipos de datos responde a dos necesidades diferentes. Los datos reales permiten comprobar si el detector opera en un escenario cercano al uso práctico. Los datos sintéticos permiten aislar variables, disponer de anotaciones completas y estudiar con más detalle el comportamiento del sistema. Esta combinación hace que los resultados no dependan únicamente de un caso idealizado ni únicamente de un caso real difícil de controlar.

Metodología.

La metodología comienza con la preparación de los tomogramas y la generación de vectores de características mediante TomoTwin. Los volúmenes se normalizan, se convierten a la convención requerida por el detector y se acompañan de coordenadas expresadas de manera consistente. A partir de las anotaciones se extraen subtomogramas que actúan como *prompts*. Cada subtomograma se codifica con TomoTwin para obtener un vector de características que condiciona la red de segmentación de ProPicker. Estos vectores no se utilizan como imágenes adicionales sino como descriptores latentes de la partícula que se desea localizar.

Una vez condicionado el modelo, la inferencia produce mapas volumétricos de localización, los cuales no son todavía la salida final del sistema. Para obtener coordenadas discretas se aplica un posprocesado que incluye umbralización, agrupamiento de respuestas cercanas y extracción de centroides. Después, las coordenadas predichas se comparan con el ground truth mediante una regla geométrica de emparejamiento. Las métricas principales son precisión, *recall* y F_1 . La precisión mide la proporción de predicciones correctas entre todas las predicciones realizadas, el *recall* mide la proporción de partículas reales que han sido recuperadas y el valor F_1 resume ambas medidas y permite evaluar el equilibrio entre omisiones y falsos positivos.

Un aspecto relevante es la implementación de la inferencia multi-*prompt*, por el cual se promedian las representaciones de varios ejemplos para reforzar los rasgos comunes de la partícula y mitigar el ruido o los errores de centrado. La agregación se realiza en el espacio latente y no directamente sobre los vóxeles. Esto permite combinar información de varios ejemplos sin tener que alinear explícitamente las partículas en el volumen. La hipótesis es que un conjunto de ejemplos bien seleccionados puede representar mejor la clase objetivo que un único subtomograma que quizá esté afectado por ruido, mala orientación o artefactos locales.

En los últimos experimentos se introduce una selección de *prompts* basada en características físicas del subtomograma y una selección en el espacio de rotaciones. Para la selección física se consideran propiedades como la concentración de energía en la zona central, la presencia de contenido estructurado y la ausencia de desplazamientos de intensidad que indiquen artefactos. Para la selección orientacional se busca cubrir poses diversas de la partícula. Dado que la tiroglobulina presenta simetría C_2 , la distancia entre orientaciones se interpreta en $SO(3)/C_2$. Esta decisión metodológica evita atribuir diferencias artificiales a poses que la propia estructura no distingue y mejora la representatividad de los ejemplos utilizados.

Resultados experimentales.

Los experimentos muestran que la inferencia basada solo en ejemplos puede identificar señal inicial, aunque suele ofrecer un rendimiento conservador. En EMPIAR-10988 el modelo base de ProPicker obtiene una precisión de 0.7049 pero un *recall* de 0.1966, con un valor F_1 de 0.3074. Este resultado indica que el modelo reconoce parte de la señal

ribosomal pero deja sin detectar muchas partículas. Tras ajustar el umbral, el equilibrio mejora ligeramente y el F_1 sube a 0.3492. Sin embargo el cambio más relevante aparece con el ajuste fino, que eleva el valor F_1 a 0.5714, demostrando ser necesario para estabilizar el sistema. La conclusión del primer experimento es que la transferencia directa contiene información útil, pero no basta para obtener un detector fiable en datos reales.

En el conjunto sintético el cambio de dominio es más severo. El modelo base prácticamente no consigue localizar la tiroglobulina y obtiene valores de F_1 cercanos a cero. Tras el entrenamiento con la división 20/5 el rendimiento medio alcanza 0.891 y cuatro de los cinco tomogramas de validación quedan por encima de 0.92. Este salto supone una mejora crítica respecto a los resultados casi nulos del modelo base en ese dominio. También muestra que ProPicker puede adaptarse de forma eficaz cuando recibe datos coherentes del objetivo. El caso más difícil es *tomo_8*, que presenta menor SNR y un rendimiento inferior al resto. Este resultado sugiere que, después del ajuste, el problema principal no es la ausencia total de señal sino la pérdida de selectividad en condiciones más ruidosas.

El análisis de eficiencia indica que el sistema alcanza buenos resultados con apenas cuatro tomogramas y que la estrategia multi-*prompt* es eficaz para reducir los falsos positivos. El experimento incremental entrena modelos con 1, 2, 4, 8, 12, 16 y 20 tomogramas manteniendo la misma validación. La curva crece con rapidez y se estabiliza después de los primeros incrementos. El mejor resultado observado aparece con 12 tomogramas e inferencia multi-*prompt*, alcanzando $F_1 = 0,919$. Lo más relevante no es solo el aumento de la métrica global, sino la reducción de falsos positivos. En un flujo real de cryo-ET esta reducción tiene valor práctico porque disminuye la necesidad de curación manual y reduce el riesgo de contaminar los subtomogramas usados en análisis posteriores.

No obstante el último experimento reveló que la elección del ejemplo sigue siendo un factor crítico. Se generó una población amplia de *prompts* a partir de tomogramas de entrenamiento y se evaluó su efecto con un mismo *checkpoint*. Los resultados muestran que muchos ejemplos mantienen un rendimiento alto pero algunos provocan una activación insuficiente del detector. El fallo principal no consiste en producir demasiados falsos positivos, sino en no activar suficiente respuesta sobre partículas reales. Los análisis no permiten explicar completamente estas caídas usando solo SNR, posición vertical, distancia al centro o hemisferio de orientación. Esto apunta a una interacción más compleja entre la calidad de la imagen, la anisotropía de la muestra, la orientación de la partícula y la representación latente generada por TomoTwin.

Interpretación conjunta.

La discusión de los resultados permite concluir que la inferencia directa puede proporcionar una señal de partida, pero el ajuste fino es una condición práctica necesaria cuando cambian los datos. El modelo base no debe entenderse como una solución cerrada, sino como una inicialización con conocimiento previo que puede reducir el coste de adaptación. Esta diferencia es importante porque evita una interpretación demasiado optimista de la universalidad. ProPicker no resuelve por sí solo todos los dominios, pero sí ofrece una base útil sobre la que entrenar con pocos datos.

La cantidad de anotación requerida es moderada y la agregación de *prompts* mejora la robustez al controlar la aparición de falsos positivos. Los resultados muestran que no siempre es necesario llegar al máximo número de tomogramas disponibles para obtener un rendimiento competitivo. A partir de cierto punto los beneficios adicionales son menores y la selección de ejemplos o la estrategia de inferencia puede ser tan importante como añadir más datos. Sin embargo, la representatividad del ejemplo seleccionado sigue influyendo

en el éxito de la detección final. Esto significa que el *prompt* no debe tratarse como un detalle secundario sino como un elemento central del detector.

Recomendaciones de optimización.

La tesis propone prioridades para un flujo de trabajo real, empezando por la consistencia en el tratamiento de los datos antes de comparar modelos. Los nombres de tomogramas, las rutas, las unidades de coordenadas, el contraste y la división entre entrenamiento y validación deben quedar fijados desde el principio. Sin esta consistencia es difícil saber si una mejora procede del modelo o de un cambio accidental en el procesamiento.

Se recomienda evitar selecciones arbitrarias de ejemplos y preferir casos estructurados y diversos, además de aplicar una política de ajuste incremental. Una estrategia razonable consiste en comenzar con pocos tomogramas, evaluar pronto y añadir datos solo cuando la validación lo justifique. También es fundamental hacer visibles los parámetros de posprocesado, ya que los umbrales, las distancias de agrupamiento y las reglas de emparejamiento forman parte del detector efectivo. Finalmente el análisis de fallos debe tratarse como una etapa estándar del proceso. Guardar mapas de localización, coordenadas, métricas por tomograma y metadatos de cada *prompt* permite identificar si un problema aparece por ruido, por mala selección del ejemplo o por limitaciones del modelo.

Limitaciones.

Aunque los experimentos analizan transferencia, ajuste fino, eficiencia de datos y robustez de *prompts*, todavía no prueban una universalidad amplia entre partículas, laboratorios y protocolos de adquisición. Además, algunos umbrales se heredan de protocolos de referencia y la robustez frente a la elección del *prompt* sigue sin resolverse por completo.

El análisis de orientaciones se estudia en detalle para la tiroglobulina, pero no se extiende a otras simetrías moleculares. Por ello, las conclusiones muestran dónde ProPicker ya es útil y dónde aún requiere mejoras, y dejan abierta la validación en conjuntos reales más amplios y en escenarios multiclase.

Conclusión.

El trabajo concluye que ProPicker es una herramienta eficaz para la localización de partículas cuando se integra en un flujo que combine el ajuste fino limitado con un control de calidad de los ejemplos. Aunque el modelo base es informativo, no se puede considerar todavía un detector universal plenamente autónomo. Su rendimiento depende del dominio, de la calidad de los datos, del ajuste fino y de la representatividad del *prompt*. Sin embargo los resultados muestran que puede transformarse en un detector sólido con una cantidad moderada de anotación y con una metodología controlada.

El avance hacia sistemas más robustos requiere incorporar mayor diversidad de datos reales, selección automática de *prompts* y representaciones que consideren explícitamente las simetrías moleculares. También será importante extender la evaluación a escenarios multiclase, estudiar aumentos compatibles con la física de adquisición y diseñar criterios que detecten *prompts* poco fiables antes de ejecutar la inferencia completa. En conjunto la tesis muestra que la universalidad no debe entenderse como ausencia total de adaptación, sino como capacidad de reutilizar conocimiento previo, reducir la supervisión necesaria y mantener un comportamiento interpretable al cambiar de dominio.

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Terms and Acronyms

Cryo-ET Cryo-electron tomography.

DL Deep Learning.

EMPIAR Electron Microscopy Public Image Archive.

Ground truth Reference annotations used as the evaluation target.

Tomogram 3D volume reconstructed from a tilt series.

Tilt series Set of projection images acquired at different tilt angles.

Missing wedge Region of unsampled Fourier space caused by the limited tilt range.

STA Subtomogram averaging.

Particle picking Identification of particle coordinates in a tomogram.

Prompt Particle-specific reference crop or embedding used to condition ProPicker.

ProPicker Promptable 3D segmentation model for particle picking in Cryo-ET.

TomoTwin Encoder used in the repository to generate prompt embeddings.

DeepETPicker Preprocessing and training framework reused by the ProPicker toolchain.

CNN Convolutional neural network.

U-Net Encoder-decoder CNN with skip connections for dense prediction.

SNR Signal-to-noise ratio.

IoU Intersection over union.

TP/FP/FN True positives, false positives and false negatives.

PCA Principal component analysis.

MRC Electron microscopy volume file format commonly used for tomograms.

SO(3) 3D rotation group.

C₂ Cyclic group of order 2, representing a 180-degree rotational symmetry.

F₁ Harmonic mean of precision and recall.

Chapter 1

Introduction

1.1. Motivation

Cryo-ET has become one of the most powerful techniques for observing macromolecular organization *in situ*. By reconstructing a 3D tomogram from a tilt series of a vitrified specimen, Cryo-ET preserves spatial context that is usually lost in purified single-particle pipelines. This advantage is especially relevant when the scientific question concerns intracellular organization, molecular crowding or the coexistence of several complexes within the same environment [1, 9].

The computational burden begins after reconstruction. Tomograms are noisy, anisotropic and large, and several downstream stages depend on knowing where the particles of interest are located. STA require coordinate estimates that are accurate enough to seed the rest of the structural analysis workflow [2, 15]. Therefore, particle picking is one of the main gates that determine whether a Cryo-ET study scales or stalls.

Manual annotation does not scale to current datasets, while classical methods such as template matching (TM) often require strong assumptions about the target structure and can become brittle when the particle changes conformation, size or local environment. These constraints motivate the search for more flexible detectors that can retain good performance under heterogeneous acquisition conditions and under low-label regimes.

This thesis is framed around that goal. It studies the current state of *universal* DL detectors for Cryo-ET localization by focusing on ProPicker, a promptable segmentation model that can either work from a prompt alone or be fine-tuned to a target particle with limited data [18]. The main value of this work lies in establishing a reproducible experimental workflow and in identifying where the current generation of promptable detectors is already useful and where it still breaks down.

1.2. Problem statement

Consider a cellular context represented as a 3D density map

$$D(\mathbf{x}) : V \subset \mathbb{R}^3 \rightarrow \mathbb{R}$$

where $\mathbf{x} \in \mathbb{R}^3$ denotes a spatial coordinate within the volume of interest V sampled by a tomogram [7]. We define prompt $\mathcal{P}$ as a compact representation of the target particle, used as example and obtained in this thesis from one or more centered subtomogram crops and their corresponding TomoTwin embeddings.

Given D and $\mathcal{P}$, the goal is to estimate a finite set of particle detections

$$\mathcal{D} = \{(\mathbf{p}_i, s_i)\}_{i=1}^N,$$

where $\mathbf{p}_i \in \mathbb{R}^3$ is the predicted particle center in voxel coordinates and $s_i \in \mathbb{R}$ is a confidence or response score derived from the predicted localization map. The detector should produce detections that match the unknown ground truth coordinate set

$$\mathcal{Y} = \{\mathbf{y}_j\}_{j=1}^M,$$

under a spatial tolerance criterion, while maintaining high precision and recall despite Cryo-ET noise, anisotropic resolution, domain shift between datasets, limited annotated tomograms and sensitivity to the chosen prompt representation.

In practice, the complexity of this problem stems from a combination of several adverse conditions:

- Very low SNR caused by dose limitations.
- Anisotropic resolution due to the missing wedge.
- Crowded cellular context.
- Limited annotations and strong cost of manual labeling.
- Computational constraints imposed by large 3D volumes.

The specific focus of this thesis is narrower than the full detection problem. The question is not whether a detector can achieve good performance on a single curated dataset under extensive supervision, but whether a *promptable* detector can remain useful when supervision is scarce, the domain changes and prompt quality is itself a variable under study.

1.3. Research questions

The work is organized around the following research questions:

1. Can prompt-only inference with ProPicker provide useful transfer on real Cryo-ET data before any specific fine-tuning?
2. How severe is the performance drop when the detector is moved to a substantially different domain?
3. How much annotated data is required to recover strong performance after fine-tuning?
4. Does aggregating several prompts improve practical robustness with respect to a single prompt?
5. Once fine-tuning is effective, is the remaining variability driven mainly by tomogram difficulty or by prompt factors such as orientation and embedding quality?

1.4. Contributions

The main contributions of the thesis are:

1. A reproducible research repository that organizes preprocessing, prompt construction, fine-tuning, inference and result export for Cryo-ET particle picking using ProPicker.
2. A real data proof of concept on EMPIAR-10988 showing that the inference with the base model is viable but insufficient for balanced performance.
3. A domain shift experiment using PolNet-synthesized thyroglobulin data demonstrating that fine-tuning is essential for maintaining model performance.
4. A data efficiency study that identifies a strong operating point with only 4 training tomograms and the best observed configuration at 12 tomograms with multi-prompt inference.
5. A prompt robustness analysis showing that residual instability is expressed mainly as prompt-dependent recall collapse and is only weakly explained by source SNR or simple rotation proxies taken in isolation.

1.5. Document structure

Chapter 2 reviews the Cryo-ET localization problem and the background needed to understand promptable detectors. Chapter 3 describes the objectives, datasets, methodology, implementation choices and evaluation protocol. Chapter 4 reports the results of the four completed experiments and discusses their implications. Chapter 5 summarizes the conclusions and outlines the future work. The appendix documents the hardware and software environment used in the project.

Chapter 2

State of the Art

This chapter is organized from general context to the specific task studied in the thesis. It first reviews how Cryo-ET data are acquired, then summarizes the main particle-picking formulations and finally focuses on promptable detectors, dataset choices and the evaluation issues that matter for the experimental chapters.

2.1. Acquisition and reconstruction workflow

Cryo-ET reconstructs a 3D volume from a series of projection images acquired at different tilt angles. Because the angular range is limited in practice, Fourier space is sampled incompletely and the reconstruction exhibits the missing wedge artifact [1, 9]. This causes direction-dependent distortions and makes pattern recognition intrinsically harder than in isotropic 3D imaging settings.

At a workflow level, the acquisition and reconstruction process can be summarized in four steps. First, a tilt series of two-dimensional (2D) projections is acquired while the specimen is mechanically tilted. Second, the images are motion-corrected and aligned, usually with fiducials or alignment models that compensate for grid deformation. Third, a 3D tomogram is reconstructed, typically through weighted or filtered backprojection or through iterative methods such as simultaneous iterative reconstruction technique (SIRT) and simultaneous algebraic reconstruction technique (SART). Fourth, the reconstructed volume is used for downstream tasks such as particle picking, segmentation or STA [11, 23]. Modern preprocessing packages such as Warp illustrate how motion correction, contrast transfer function (CTF) handling and reconstruction-adjacent preparation have become central parts of cryo-electron microscopy (Cryo-EM) and Cryo-ET workflows [16]. This sequence matters because the localization problem studied in this thesis is not applied to raw particle images but to reconstructed volumes that already contain the acquisition footprint of the microscope and the reconstruction algorithm.

Figure 2.1 provides an overview of the full Cryo-ET pipeline discussed in this chapter, from tilt-series acquisition to tomogram reconstruction, denoised slices and the different levels of difficulty encountered during particle picking.

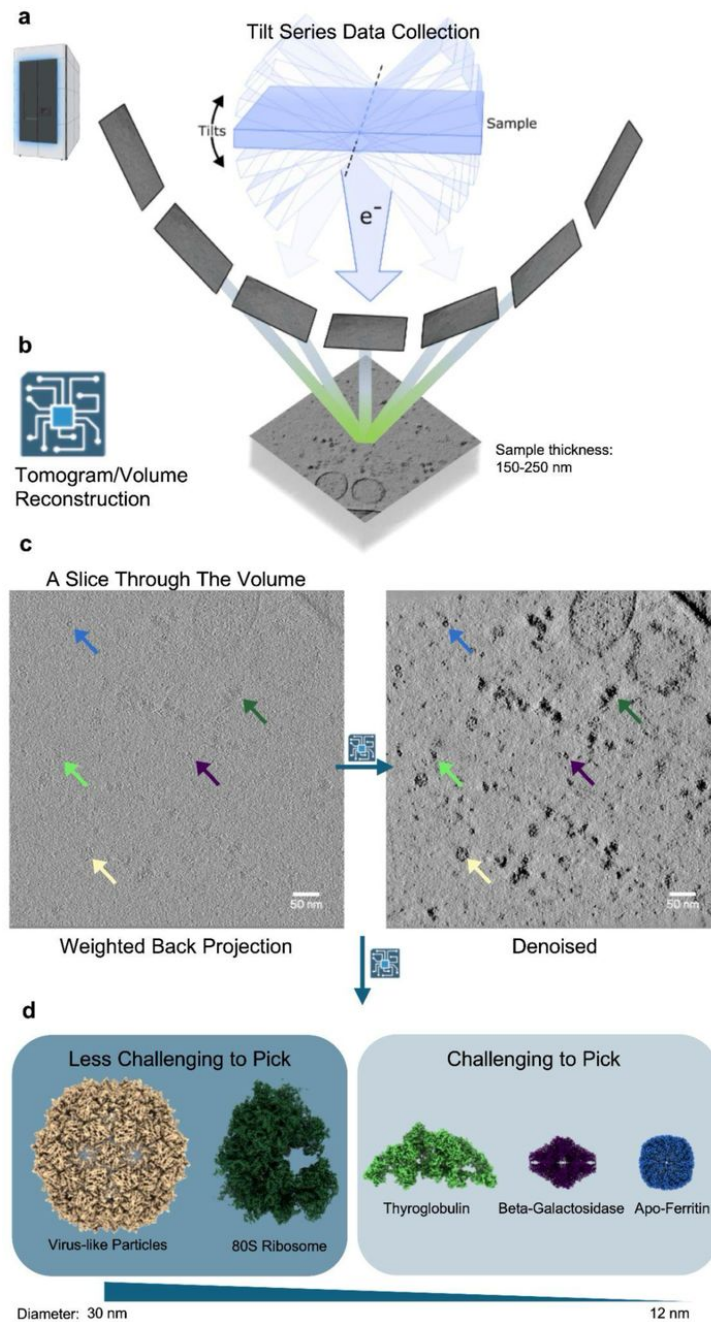


Figure 2.1: Overview of the Cryo-ET workflow from tilt-series acquisition to tomogram reconstruction and denoised slices, together with an illustrative comparison of particle classes that are easier or harder to localize automatically. (a) Tilt-series acquisition as the specimen is imaged over multiple angles. (b) Tomogram reconstruction from those projections. (c) Comparison between a weighted-back-projection slice and a denoised slice, where arrows indicate visible particles. (d) Particle classes with different picking difficulty, from larger virus-like particles and ribosomes to smaller targets such as thyroglobulin, β -galactosidase and apoferritin. Adapted from Figure 1 of Peck et al. [12].

2.2. Macromolecule localization and subtomogram analysis

Particle picking seeks to estimate the coordinates of particle centers inside the tomogram. Those coordinates are then used to extract subtomograms, which can be aligned, classified and averaged in order to recover a higher-SNR structural signal [2, 15]. A weak picker damages the entire downstream workflow: if recall is low, particles are missed; if precision is low, later stages waste time on FP or degrade due to contamination.

2.2.1. Classical localization strategies

Traditional Cryo-ET localization methods fall roughly into two groups. TM compares the tomogram with a reference template across positions and orientations. Feature-based methods rely on filters, local maxima and hand-designed rules. TM remains the most widely accepted method, but it can be computationally expensive and it assumes that a sufficiently representative template is available. Hand-crafted strategies are usually more brittle when the target varies in appearance or when background clutter changes substantially.

2.2.2. Deep Learning formulations for 3D particle picking

Modern learning-based approaches usually reframe particle picking as one of the following problems:

- Voxel-wise segmentation or center heatmap prediction.
- 3D object detection on patches or subvolumes.
- Similarity-driven retrieval and matching in embedding space.

Architectures inspired by U-Net remain common because they combine local detail with contextual information and can be adapted to volumetric inputs [14]. DeepFinder is a representative Cryo-ET example of this strategy: it uses a 3D convolutional segmentation network to produce class-wise score maps, then converts the segmentation output into particle coordinates through clustering [10]. More general object detection literature distinguishes anchor-based and anchor-free formulations [6, 13, 17, 22]. In Cryo-ET, however, the representation of interest is often a particle center rather than a tight 3D bounding box, so segmentation maps and center-localization maps are especially natural.

Figure 2.2 shows the CNN architecture used by DeepFinder together with its downstream localization workflow. The network follows an encoder-decoder design where convolutional blocks extract volumetric features, max-pooling increases the receptive field, up-sampling restores spatial resolution and skip concatenations transfer fine-scale information to the decoder. The trained CNN converts an input tomogram into a multiclass segmentation map, after which clustering turns connected segmented regions into particle coordinates.

Despite the progress of generic 2D detectors [5], Cryo-ET retains a clear volumetric character. Working with 3D patches is therefore advantageous, but it also imposes memory constraints that force careful tiling, padding and response aggregation.

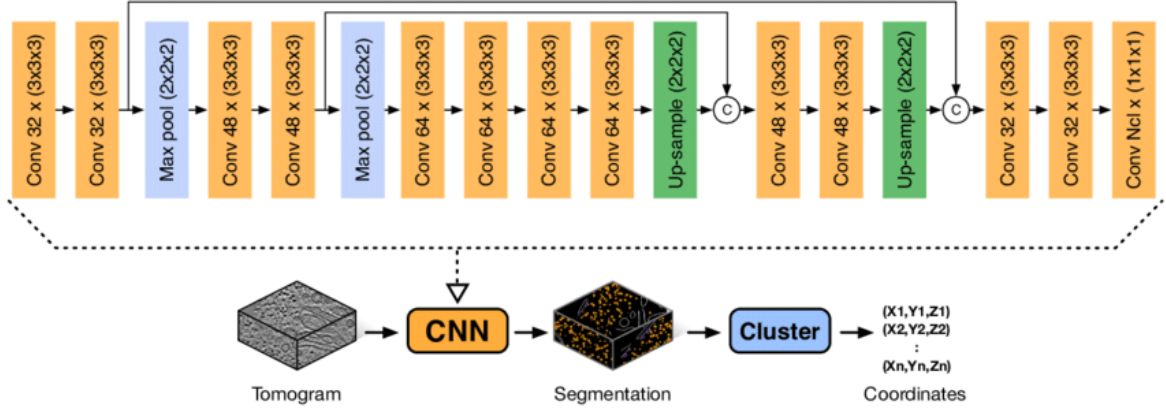


Figure 2.2: CNN architecture and localization workflow used in DeepFinder. Orange blocks denote 3D convolutional layers, blue blocks denote max-pooling operations, green blocks denote up-sampling operations, and circled C nodes denote concatenation of encoder and decoder feature maps. The final N_c -channel convolution produces a multiclass segmentation map; clustering then converts segmented regions into particle coordinates. Adapted from Extended Data Figure 1c of Moebel et al. [10].

2.2.3. Promptable detectors and the idea of universality

The notion of a universal detector in Cryo-ET is closely tied to flexibility. A model that performs very well on one fixed class but requires retraining from scratch for every new particle is useful, but not universal in any practical sense. Promptable detectors attempt to bridge that gap by conditioning a general model on a reference example of the target particle.

ProPicker is one of the most recent examples of this family. It combines a prompt encoder with a 3D segmentation network that is conditioned on prompt-derived features and then produces a localization map from which coordinates can be extracted [18]. The main attraction of this design is twofold. First, prompt-only inference enables flexible targeting of new particles without retraining. Second, if that transfer is insufficient, fine-tuning can specialize the same model with comparatively little data.

Figure 2.3 summarizes that workflow. A prompt crop is encoded into a particle-specific representation, the tomogram is processed by a conditioned segmentation model, and the final particle coordinates are recovered from the predicted segmentation mask through post-processing.

This design is conceptually aligned with broader trends in promptable vision systems, but Cryo-ET adds constraints that make the problem harder. A prompt is not always equally informative. Its orientation, local context, centering quality and embedding statistics may all affect the downstream response of the segmentation model. Therefore, universality is not only about training data diversity or network capacity; it is also about the stability of the conditioning mechanism.

2.2.4. From tilt-series acquisition to rotation-sensitive prompts

In an ideal isotropic 3D imaging system, rotating a particle would mostly produce a rotated copy of the same signal. Cryo-ET breaks that symmetry because the specimen is scanned over a limited angular range and around a single mechanical tilt axis. The resulting missing wedge removes direction-dependent Fourier information and creates

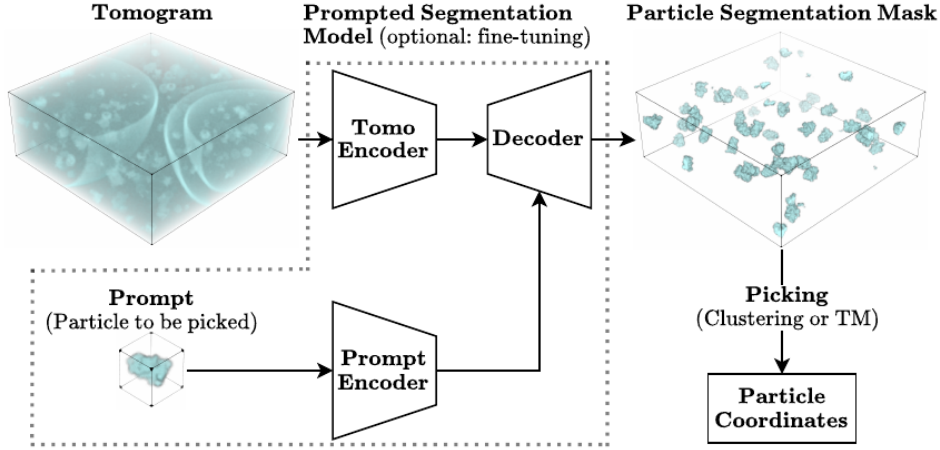


Figure 2.3: Overview of the ProPicker workflow. A prompt encoder extracts features from a reference particle, those features condition the tomogram segmentation model and the predicted segmentation mask is converted into particle coordinates by a downstream picking step. Adapted from Figure 1 of Wiedemann et al. [18].

anisotropic blur and elongation in the reconstructed tomogram [1, 9]. Consequently, two subtomograms containing the same macromolecule at different orientations are not related by a clean rigid rotation alone.

This can be written schematically as

$$y \approx \mathcal{A}_{\text{mw}}(Rx) + \eta,$$

where x is the underlying particle, R is its rigid orientation, η is noise and $\mathcal{A}_{\text{mw}}$ is the acquisition-reconstruction operator induced by limited-angle tomography. The key point is that $\mathcal{A}_{\text{mw}}$ is not rotationally neutral: rotating the object before reconstruction is not equivalent to rigidly rotating the reconstructed volume. This anisotropy is precisely why missing-wedge compensation methods such as IsoNet are needed [8].

This matters directly for promptable detection. A prompt crop contains not only the target particle, but also the missing-wedge footprint, local reconstruction context, possible centering error and residual contrast distortions. A geometrically favorable prompt can therefore yield a more stable representation than another prompt from the same class at a less favorable orientation.

The model architecture adds a second difficulty. Standard CNNs are naturally equivariant to translations, but not to arbitrary rotations [3]; in 3D, rotation equivariance requires specialized constructions such as steerable convolutions [19]. ProPicker gains flexibility by conditioning the detector on a prompt, but that does not make the overall pipeline rotation-invariant [18]. Experiment 4 (EXP4) should therefore be read as a targeted test of the full acquisition–reconstruction–prompt–model chain.

The specific target used in the synthetic dataset adds another source of rotational ambiguity. Human thyroglobulin is a homodimer with global C_2 symmetry, and the reported cryo-EM structure is approximately $250 \times 160 \times 110$ Å, with the shortest dimension aligned with the C_2 axis [24]. Two poses that differ by a 180° rotation around that axis are physically equivalent, so a pose-sensitive system naturally operates over $SO(3)/C_2$ rather than pure $SO(3)$.

For thyroglobulin, the problematic rotations are therefore of two kinds: symmetry-

equivalent rotations, which should ideally be collapsed by the loss or evaluation metric, and acquisition-amplified ambiguities, where discriminative pose differences are degraded by the missing wedge. This is why physically plausible 3D augmentations, symmetry-aware losses and matched missing-wedge training are attractive strategies in Cryo-ET [8, 19, 23].

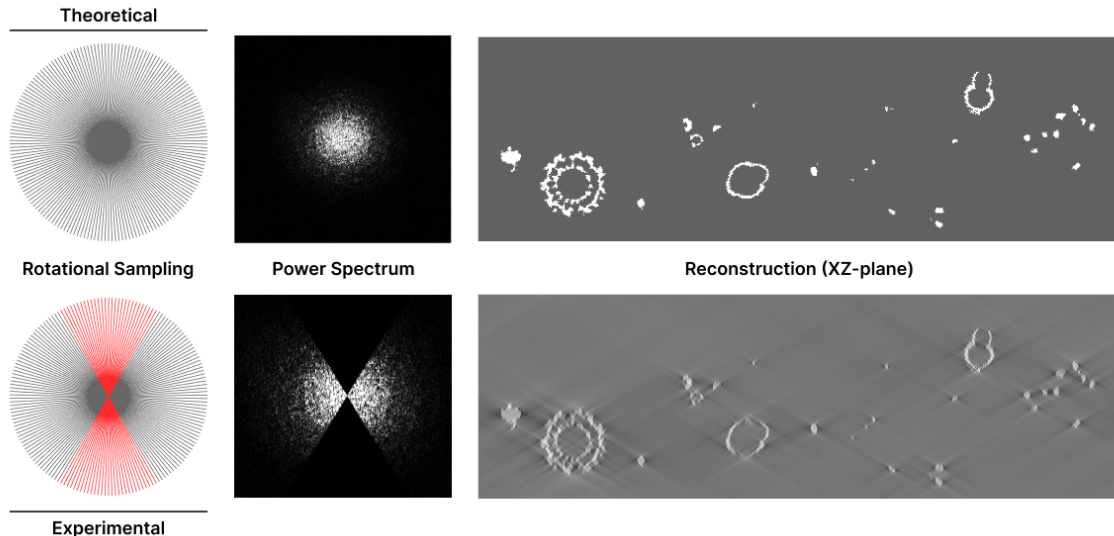


Figure 2.4: Wedge-like reconstruction problem in Cryo-ET. The theoretical case shows complete rotational sampling and a reconstruction without missing angular information, while the experimental case shows limited sampling, a missing wedge in Fourier space and elongation artifacts in the reconstructed XZ plane. In the experimental rotational-sampling panel, the red angular sectors indicate directions that are not acquired because the specimen cannot be tilted through a full 180° range; these unsampled directions correspond to the dark wedge-shaped gaps in the power spectrum. This acquisition-level anisotropy explains why rotated subtomograms are not simple rigid copies of one another and why prompt orientation can affect downstream particle picking. Adapted from the CryoET Data Portal documentation [4].

2.3. Real and synthetic datasets in Cryo-ET

Real and synthetic tomograms serve complementary purposes. Real datasets capture the complexity of laboratory conditions, reconstruction artifacts and biological variation. Synthetic datasets, by contrast, provide full ground truth and allow controlled experiments on noise, density, class distribution and missing-wedge-related distortions. A serious evaluation program benefits from both.

That is precisely why this thesis combines EMPIAR-10988 and PolNet synthetic data. The real dataset answers whether the detector is useful in a realistic scenario. The synthetic dataset answers more controlled questions about adaptation, data efficiency and prompt robustness, because it allows stable train/validation splits and exhaustive metric computation.

Chapter 3

Objectives and Methodology

This chapter joins the experimental objectives with the methodological decisions that make the later results comparable. The goal is to describe not only what was evaluated, but also how the repository enforces stable data splits, coordinate conventions, prompt construction, training schedules, inference outputs and evaluation rules. This organization is important because the thesis studies small practical differences, such as prompt aggregation and training-set size, whose interpretation would be weak if preprocessing or metrics changed between runs.

3.1. Objectives

The thesis investigates the practical value of ProPicker as a foundation for universal macromolecule localization in Cryo-ET. Rather than focusing on optimizing leaderboard metrics, the project objective is to study the current limits of ProPicker and translate these findings into a concrete optimized pipeline. This approach aims to maximize the model’s performance through systematic experimentation without altering the architecture.

The specific objectives are:

1. Build a reproducible repository that can support repeated fine-tuning and inference studies across several datasets.
2. Validate the complete workflow on real Cryo-ET data.
3. Quantify domain-shift effects on a synthetic dataset with full ground truth.
4. Measure data efficiency and compare prompt aggregation strategies.
5. Study the robustness limits of the current prompt-conditioning mechanism.
6. Establish a project foundation that scales easily to new experiments in future work.

3.2. Datasets

3.2.1. Experimental data: EMPIAR-10988

The real dataset uses the EMPIAR-10988 ribosome case already supported by the ProPicker examples. In the repository, the experiment focuses on the *cyto_ribosome* target. Training is performed on tomogram *TS_029* and validation on *TS_030*. The

workflow uses a centered crop with *crop_delta=64*, which yields a controlled subvolume of $128 \times 128 \times 128$ voxels around the region of interest.

This setup is a proof of concept on real data, not a full-scale evaluation over the entire EMPIAR collection. The primary goal was to verify the pipeline, from prompt extraction to fine-tuned inference, under realistic data conditions while keeping the experiment small enough for careful debugging. Furthermore, cytoplasmic ribosomes were selected as the initial target because they are relatively easy to pick. Their large size, distinctive symmetrical form and high contrast relative to the background make them ideal for validating new localization pipelines.

Because Experiment 1 (EXP1) focuses on cytoplasmic ribosomes, Figure 3.1(a) is included as a structural reference for the target macromolecule. It is illustrative rather than dataset-derived, but it helps anchor the discussion around the particle class being localized in the tomograms.

3.2.2. Synthetic dataset: PolNet

The main synthetic dataset consists of PolNet synthetic tomograms. The completed experiments focus on the *thyroglobulin* class, corresponding to label 7 in the comma-separated values (CSV) annotations. The repository-level reports document 25 tomograms and approximately 3327 labeled thyroglobulin particles in total. The split used from Experiment 2 (EXP2) onward is fixed:

- Validation subset: *tomo_5*, *tomo_6*, *tomo_7*, *tomo_8* and *tomo_9*.
- Training subset: the remaining 20 tomograms.

Keeping the validation set fixed is essential because EXP2, Experiment 3 (EXP3) and EXP4 are intended to be directly comparable.

Figure 3.1(b) is relevant for the synthetic dataset mainly because it shows the global form of thyroglobulin. The particle is large, elongated and not close to spherical, so its projection and reconstructed appearance can change strongly with orientation. Under limited-angle Cryo-ET acquisition, the missing wedge can therefore remove different parts of the structural signal depending on how the particle is rotated in the tomogram. This makes the target a useful case for studying form ambiguity, prompt orientation effects and rotation sensitive failures.

3.3. Method overview

3.3.1. Adopted detector and design rationale

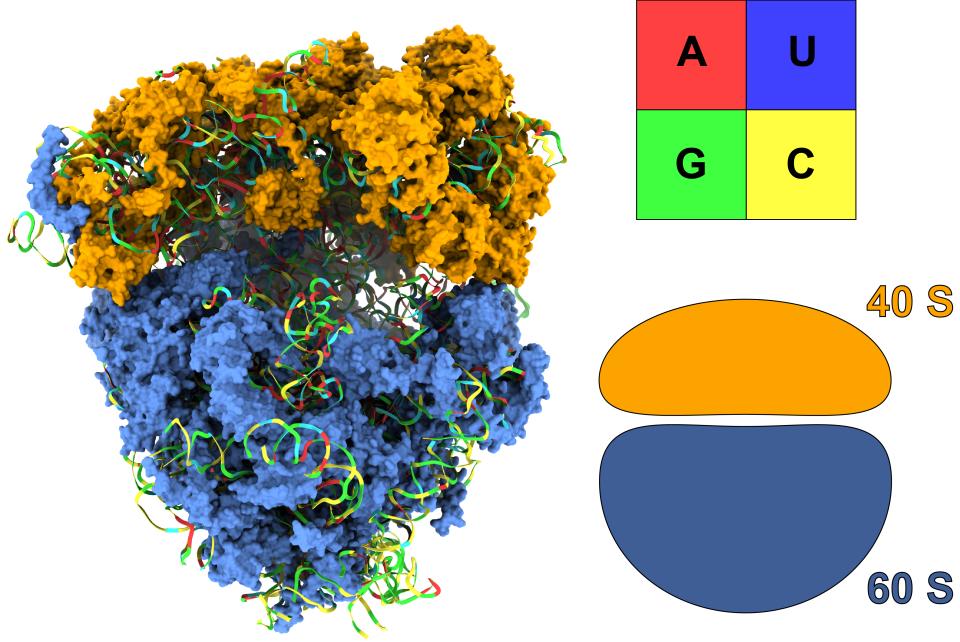
The detector used throughout the thesis is ProPicker [18]. The choice is motivated by its promptable design and by its ability to support two complementary modes:

- **Prompt-only inference**, which tests how much cross-domain transfer can be obtained without task-specific training.
- **Particle-specific fine-tuning**, which tests how efficiently the same model can adapt when transfer alone is insufficient.

This makes ProPicker a suitable tool for studying universality, as it allows us to systematically evaluate how far a single detector can generalize across particle types through prompting alone, and how efficiently it can adapt when fine-tuning is required.

(a)

80S human ribosome



(b)

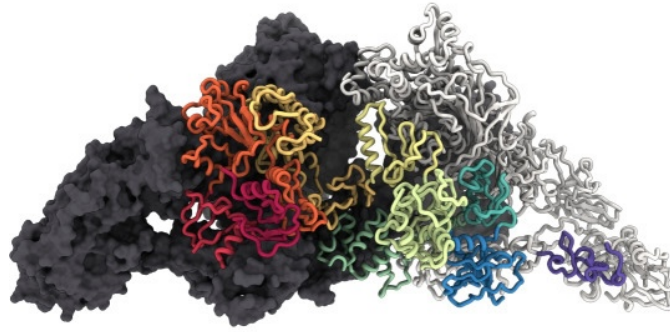


Figure 3.1: Structural references for the two particle targets used in the thesis. (a) Illustrative 80S human ribosome model used as a visual reference for the EMPIAR-10988 cytoplasmic-ribosome experiment. The information on the right summarizes the ribosome composition: the colored *A*, *U*, *G* and *C* boxes identify the four RNA bases, and the 40S and 60S schematics indicate the small and large eukaryotic ribosomal subunits. Reproduced from Ykust [20], licensed under Creative Commons Attribution (CC BY) 4.0. (b) Type 1 repeats in the human thyroglobulin structure. One human thyroglobulin (hTG) monomer is shown as a charcoal surface, while the other monomer highlights the individual type 1 repeats as colored ribbons. Adapted from Figure 2 of Adaixo et al. [24].

3.3.2. Prompt construction and inference

Each prompt is built from a subtomogram centered on a labeled particle. The prompt encoder transforms the crop into an embedding that conditions the segmentation network. Inference then produces a localization map over the target tomogram, after which clustering and thresholding convert the response into discrete coordinates.

The thesis studies three prompt regimes:

1. One fixed prompt per experiment, used as the simplest baseline.
2. A multi-prompt embedding obtained from several instances to stabilize inference.
3. Rotation-diverse prompt sets, used to probe orientation sensitivity.

Prompt embedding with TomoTwin. In this workflow, an embedding is a compact numerical vector that represents the visual and structural content of a prompt subtomogram. TomoTwin produces this vector by passing the $37 \times 37 \times 37$ crop through a pretrained 3D encoder and taking the encoder output as the prompt descriptor. The result is no longer a voxel volume, but a latent representation that ProPicker can use to condition its segmentation network toward the particle class shown by the prompt.

The implementation became progressively more selective as the investigation advanced. In EXP1 and EXP2 the prompt is a fixed, manually validated positive instance, because the objective there is to measure transfer and domain adaptation, not prompt search. In EXP3 the study introduces a robust prompt-ranking function in order to avoid a naive strategy based only on local variance. The physical-quality score used in that selector is

$$q_i = w_f \log(f_i) + w_c \log(c_i) - w_{dc} d_i,$$

where f_i is a mid-frequency to high-frequency Fourier power ratio, c_i is a center-to-border intensity ratio and d_i is a direct-current (DC) penalty term that penalizes offset-dominated crops. The default weights are $w_f = 1.0$, $w_c = 0.9$ and $w_{dc} = 0.6$. The selector first ranks physically plausible candidates, keeps the best $k = 8$, and can in principle be extended with an embedding-consensus term. In the actual EXP3 single-prompt workflow, the final prompt was selected from the physical score alone.

The multi-prompt branch of EXP3 follows a different logic. It extracts ten candidate subtomograms, computes their individual embeddings with TomoTwin and represents the particle class in two ways: through the first embedding as a single-instance prompt and through the arithmetic mean of all ten embeddings as a multi-instance prompt. The selection stage does not simply take the ten most variant crops. Instead, it sorts candidates by variance and samples uniformly across that ranked list, which preserves diversity while avoiding a collapse onto extreme outliers.

Formation of the multi-prompt embedding

The multi-prompt is formed before inference, from the already extracted $37 \times 37 \times 37$ prompt subtomograms. Let $\{s_i\}_{i=1}^M$ be the available candidate crops for the target particle class. Each s_i is still a real 3D crop centered on one annotated thyroglobulin particle. The candidates are first ranked by local intensity variance, but the final set is not chosen as the top ten by that score. Instead, ten indices are sampled uniformly across the variance-ranked list. This keeps prompts with different contrast and structural strength while reducing the risk that the representation is dominated by a few noisy or unusually high-variance crops.

Each selected crop s_i is then passed independently through the TomoTwin prompt encoder,

$$e_i = \text{TomoTwin}(s_i),$$

which produces one embedding vector per selected particle instance. At this point the information is no longer represented as a visible subtomogram. The crop has been mapped into a latent feature space where nearby vectors should correspond to particles with similar structural appearance. The «mixture» of the ten thyroglobulin examples is therefore not performed in voxel space and does not create a new averaged 3D particle volume.

The operational multi-prompt representation is instead the arithmetic mean of the ten instance-level embeddings,

$$e_{\text{multi}} = \frac{1}{N} \sum_{i=1}^N e_i, \quad N = 10.$$

This averaged vector is saved in the same JavaScript Object Notation (JSON) format expected by ProPicker, under the thyroglobulin class key. Therefore the detector still receives a single conditioning input at inference time, but that input is a consensus feature representation of several real particles rather than one isolated crop. It can be interpreted operationally as a more representative prompt for the particle class: common thyroglobulin features are reinforced, while instance-specific effects such as noise, centering error, local background and unusual orientation are partially smoothed out. The single-instance prompt used for comparison is saved from the first selected embedding, while the full set of individual embeddings is kept separately for analysis.

EXP4 adds a third layer of control by selecting prompts in quaternion space. After quality filtering, prompts are chosen greedily to maximize the minimum angular separation

$$\theta(q_i, q_j) = 2 \arccos(|q_i \cdot q_j|),$$

with a target minimum distance of 30° . This is the operational definition of rotational diversity used throughout the robustness study.

3.3.3. Preprocessing and annotation strategy

The preprocessing workflow is deliberately conservative because the thesis aims to compare experimental factors rather than to re-optimize the entire toolchain at every exploratory stage. The shared pipeline can be summarized as follows:

1. Input tomograms are read in MRC format and standardized into a common experimental layout.
2. Contrast is inverted when needed so that the synthetic and real crops follow the ProPicker/TomoTwin convention of bright particles on dark background.
3. Particle coordinates are expressed in the coordinate convention required by the detector and the evaluator.
4. Prompt subtomograms of $37 \times 37 \times 37$ voxels are extracted around labeled particles.
5. Prompt embeddings are generated with the TomoTwin encoder bundled with the repository workflow.

6. The preprocessing stage produces normalized tomograms, labels and train/test definitions that are reused consistently in fine-tuning and inference.

For experiments that use the PolNet data, the pipeline contains two additional transformations that are scientifically important. First, the annotation metadata are reconciled with the local identity of each tomogram so that the dataset remains internally consistent. Second, the thyroglobulin coordinates are converted from Å to voxel units using the voxel-size metadata read from the tomograms. This solves the common error of comparing voxel-based predictions with physical coordinate annotations, a major hurdle for pipeline reproducibility.

3.3.4. Fine-tuning pipeline

Fine-tuning follows the standardized ProPicker training protocol used throughout the study. Across the PolNet experiments, the main shared settings are:

- Patch size 72.
- Padding size 12.
- Label diameter 21.
- Learning rate 10^{-3} .
- Batch size 2 for the synthetic experiments.

EXP1 differs slightly because it follows the ribosome proof-of-concept regime, including a batch size of 8 and 75 epochs for the centered 128^3 crop. In the synthetic study, the incremental schedule of EXP3 adapts the number of epochs to the number of training tomograms in order to avoid unfairly undertraining the smallest subsets. The implemented schedule is 100, 80, 50, 35, 25, 22, and 20 epochs for training sets of 1, 2, 4, 8, 12, 16, and 20 tomograms respectively.

An important methodological detail is that all long runs are defined from the same preprocessing products, the same split definitions and the same evaluation rules before training starts. This is not cosmetic: it removes path-dependent or setup-dependent variation and ensures that differences between experiments can be interpreted as scientific differences rather than procedural ones.

3.3.5. Monitoring, exports and visualization

The framework supports three complementary inspection levels: scalar metrics for comparison, full localization maps for qualitative review and prompt metadata tables for later robustness analysis. Localization maps can be represented both as tensor outputs for numerical post-processing and as MRC volumes for volumetric inspection in ParaView. This separation between scalar, volumetric and prompt-level evidence is important because no single inspection level is sufficient on its own to explain detector behaviour in Cryo-ET.

For this reason, training and inference runs are not treated as disposable executions. Checkpoints, prompt embeddings, predicted coordinates, localization maps and summary tables are stored by experiment and configuration. This is especially important in EXP3 and EXP4, where several training increments, prompt variants and candidate model states have to remain auditable under the same validation protocol.

3.4. Experimental protocol

3.4.1. Scientific workflow and reproducibility controls

The scientific workflow used in the thesis follows the same sequence in all four experiments:

1. Formulate a narrowly defined experimental question and an operational hypothesis.
2. Freeze the relevant split, preprocessing rules and metric implementation before launching long runs.
3. Isolate the factor under study so that only one main axis changes at a time.
4. Execute training or inference under standardized conditions that save checkpoints, configurations and outputs in deterministic result sets.
5. Perform quantitative evaluation first and qualitative or failure analysis second.

This order is central to the credibility of the results. EXP1 changes model adaptation and inference threshold, not the dataset. EXP2 changes domain while keeping the new PolNet split fixed. EXP3 changes training-set size and prompt aggregation while preserving the split and the downstream evaluator. EXP4 changes prompt orientation and checkpoint family while reusing the same validation tomograms. The framework therefore acts not only as storage, but as an experimental control mechanism.

3.4.2. Baselines and controlled comparisons

The experimental program is cumulative. Table 3.1 summarizes the completed experiments and the main comparison made in each one.

Exp.	Dataset	Goal	Main comparison
EXP1	EMPIAR-10988 ribosomes	Validate the full workflow on real data	Base ProPicker vs threshold-tuned base vs fine-tuned model
EXP2	PolNet synthetic, thyroglobulin	Measure severe domain shift and recovery through adaptation	Base ProPicker vs fine-tuned model on a fixed 20/5 split
EXP3	PolNet synthetic, thyroglobulin	Measure data efficiency and prompt aggregation effects	Incremental training sizes and single-prompt vs multi-prompt inference
EXP4	PolNet synthetic, thyroglobulin	Study residual robustness after fine-tuning	Base vs fine-tuned families and prompt-level rotational variability

Table 3.1: Overview of the completed experiments.

3.4.3. Metrics

The main scalar metrics are

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

where TP , FP and FN denote correct detections, spurious detections and missed particles, respectively. Precision measures the fraction of reported detections that are correct and is therefore useful for quantifying contamination of the picked subtomogram set, whereas recall measures the fraction of real particles that are recovered and is therefore useful for quantifying how much potentially valuable structural signal is being lost. The F_1 score is the harmonic mean of precision and recall, so it increases only when the detector maintains a reasonably balanced operating point instead of improving one term while collapsing the other.

In Cryo-ET particle picking, F_1 is commonly used as the main summary metric because the task is strongly imbalanced and because both kinds of error are costly for downstream analysis: too many FP waste inspection effort and contaminate STA, while too many FN reduce the amount of recoverable structural information [2, 15]. For that reason, a global accuracy measure would be misleading, whereas F_1 provides a compact description of practical detector quality while precision and recall remain available to indicate the dominant failure mode.

These metrics are inseparable from post-processing, because clustering, thresholding, coordinate extraction, geometric matching and size or overlap filters all influence the final counts of TP, FP and FN.

3.4.4. Hyperparameter policy

The objective of the thesis is not to perform a broad hyperparameter search. Once a stable setup was found, the policy was to keep it fixed whenever possible and only change one axis of variation at a time. This decision is especially important for EXP3 and EXP4, where the point is to compare training set size and prompt properties, not to co-optimize all training settings jointly.

Parameters like patch size, padding, prompt size, contrast inversion and the PolNet validation split are kept constant once they become part of the shared synthetic dataset. Only the factors required by the research question like threshold calibration in EXP1, domain adaptation in EXP2, sample efficiency and prompt aggregation in EXP3 and rotational prompt diversity in EXP4 are allowed to move.

Chapter 4

Experiments, Results and Discussion

This chapter documents the four experiments conducted to answer the hypotheses posed in this thesis. The first experiment tests the model’s performance with real data from EMPIAR-10988 ribosomes, the second measures synthetic domain displacement in PolNet thyroglobulin, the third studies the efficiency and learning capacity of ProPicker and the fourth investigates the model’s weaknesses based on the results obtained. Each section corresponds to a single experiment, is named according to the result obtained and follows the same internal structure: question and hypothesis, experiment summary, results, interpretation, and conclusions.

4.1. Fine-tuning turns real-data transfer into balanced ribosome detection

Question and hypothesis. The first question is whether ProPicker contains enough transferable prior information to detect ribosomes on real Cryo-ET data before any adjustment. The hypothesis is that the pretrained detector should show useful signal, but that this signal will be poorly calibrated and biased toward precision rather than recall. The expected result is therefore not immediate deployment quality, but evidence that lightweight fine-tuning can convert a conservative detector into a balanced one.

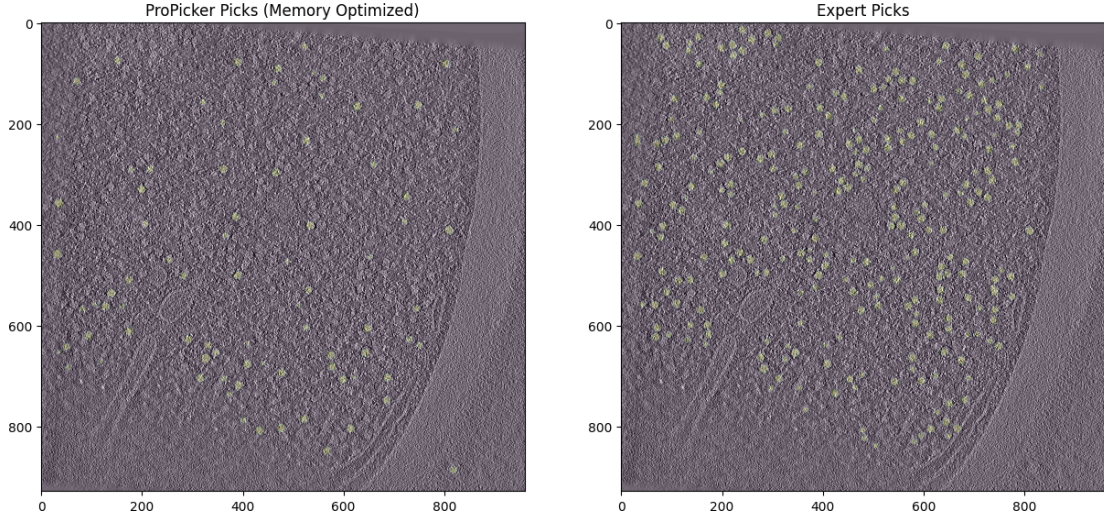
Experiment summary. The experiment uses the *cyto_ribosome* case of EMPIAR-10988, with *TS_029* for training and *TS_030* for validation under *crop_delta=64*. The methodology is carried out in two stages. First, the base model is run in prompt-only mode and its localization map is transformed into coordinates through cluster post-processing. It calibrates:

- The binarization threshold, which controls the precision-recall balance through the size of the activated clusters.
- The minimum cluster size, initialized near 0.1 times the expected ribosome ball volume, and the maximum cluster size, initialized near 1.2 times the same volume.

The evaluator then applies the same criterion used in the reference workflow, counting a detection as correct when it reaches $\text{IoU} \geq 0.6$. Second, the model is fine-tuned on the cropped ribosome setting with the same prompt construction and the same evaluation rule. This preserves the scientific comparison: only the model adaptation changes, not the data split or the evaluator.

Results. Figure 4.1 shows the key finding of the real-data experiment. The pretrained model is not useless; it is conservative. It finds ribosomes with decent precision, but its recall is too low for operational use. Threshold tuning improves F_1 slightly, which confirms that part of the issue is calibration. Still, the decisive improvement comes from fine-tuning: the F_1 score rises from the 0.31 to 0.35 range into 0.5714, with recall more than doubling relative to the base model.

(a)



(b)

Setting	Precision	Recall	F_1	TP	FP	FN
Base ProPicker	0.7049	0.1966	0.3074	547	229	2236
Base with optimized threshold	0.5125	0.2648	0.3492	737	701	2046
Fine-tuned model	0.5677	0.5752	0.5714	1044	795	771

Figure 4.1: EXP1 fine-tuning effect on real EMPIAR-10988 ribosome data. (a) Qualitative comparison between detector picks and expert annotations on the same tomographic slice. The detector localizes ribosome-like regions, but the denser annotation panel explains why recall is the limiting factor before adaptation. (b) Quantitative comparison between the base model, the threshold-tuned base model and the fine-tuned model. Threshold tuning changes calibration and raises F_1 only modestly, whereas fine-tuning raises recall from 0.1966 to 0.5752 while preserving usable precision.

Interpretation and conclusions. The result justifies the rest of the thesis. First, the promptable formulation carries useful information even on real data. Second, threshold calibration matters enough to deserve explicit treatment. Third, calibration alone does not close the gap: the real change of regime comes from fine-tuning, which lifts recall from 0.1966 to 0.5752 and reduces the false-negative count from 2236 to 771. Figure 4.1 shows the same conclusion in both quantitative and spatial form: the detector finds plausible particles, but it undersamples the expert annotation field until the model is adapted. In scientific terms, prompt-only transfer is informative, but insufficient for balanced particle picking on real ribosome data.

4.2. Fine-tuning is required under synthetic domain shift

Question and hypothesis. The next question is whether the signal observed in EXP1 survives a much stronger domain shift. The hypothesis is that the pretrained model will collapse on PolNet synthetic thyroglobulin, but that fine-tuning on a controlled split should recover high performance if the rest of the pipeline is coherent. The expected result is therefore a sharp separation between prompt-only transfer and domain-adapted inference.

Experiment summary. The experiment freezes the now-standard 20/5 split of the 25 PolNet tomograms and keeps that validation set unchanged in all later synthetic experiments. The preprocessing stage performs the following transformations in a reproducible order:

1. Reconcile annotation metadata with the local identity of each tomogram.
2. Retain only label 7, corresponding to *thyroglobulin*.
3. Convert particle coordinates from Å to voxels.
4. Extract $37 \times 37 \times 37$ prompts.
5. Invert contrast to match the ProPicker/TomoTwin convention.
6. Generate the normalized inputs required by the detector.

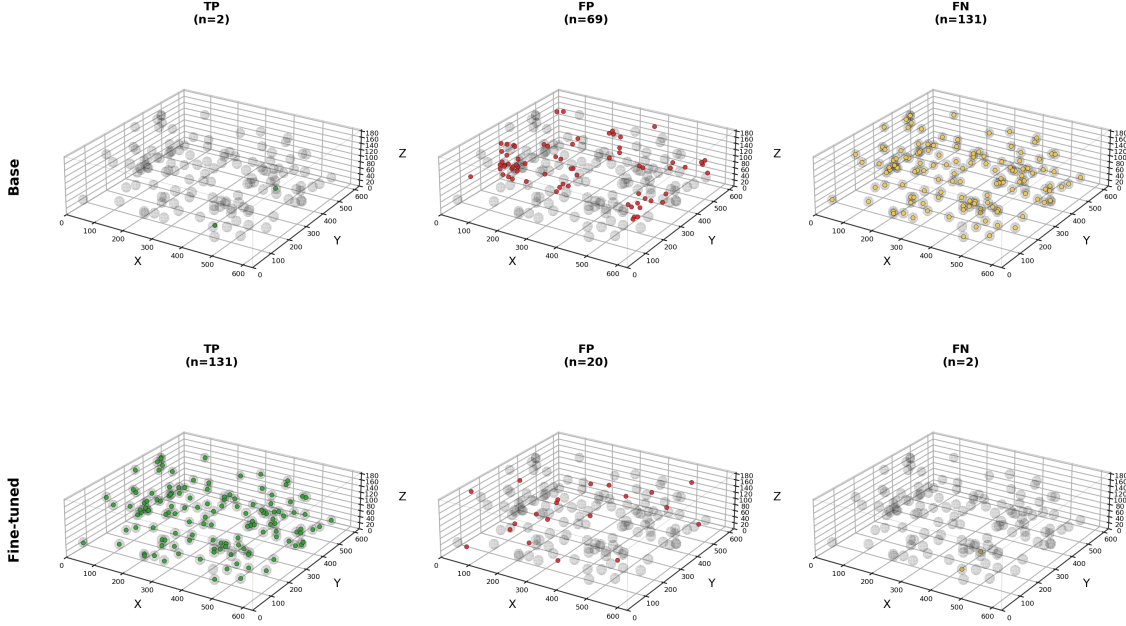
The evaluation stage uses the same validation tomograms in all comparisons and matches predicted centers to ground truth with a threshold of half the prompt size. This is a simple but crucial control: the experiment compares models under a fixed geometric rule rather than under manually adjusted tolerances.

Results. The contrast in Figure 4.2(b) is the strongest result of the thesis. The pretrained model essentially fails on this domain, with mean precision, recall and F_1 all close to zero. In practice, the base detector does not transfer from its original training distribution to PolNet thyroglobulin: it produces too few useful matches and leaves almost the complete validation particle field undetected.

After fine-tuning on the 20 training tomograms, the same pipeline changes regime. The adapted model reaches mean precision 0.822, mean recall 0.983 and mean $F_1 = 0.891$ on the fixed five-tomogram validation subset. The high recall is especially important because it shows that the model has learned to activate on the new synthetic particle class rather than only improving confidence calibration. The remaining errors are not distributed uniformly across the validation set. Four tomograms reach F_1 values around 0.92–0.94, while *tomo_8* drops to $F_1 = 0.706$ because its precision falls to 0.565 despite recall remaining high at 0.939.

The representative spatial comparison in Figure 4.2(a) complements the table by showing how the fine-tuned model recovers the particle distribution that the base model mostly misses.

(a)



(b)

Validation tomogram	Precision	Recall	F_1	TP	FP	FN
<i>tomo_5</i>	0.894	1.000	0.944	127	15	0
<i>tomo_6</i>	0.895	0.992	0.941	119	14	1
<i>tomo_7</i>	0.868	0.985	0.923	131	20	2
<i>tomo_8</i>	0.565	0.939	0.706	108	83	7
<i>tomo_9</i>	0.888	1.000	0.941	127	16	0
Mean fine-tuned performance	0.822	0.983	0.891	—	—	—
Mean base performance	0.009	0.008	0.008	—	—	—

Figure 4.2: EXP2 domain-shift recovery on the fixed PolNet validation split. (a) Three-dimensional TP, FP and FN distributions for *tomo_7* before and after fine-tuning. The base model produces very few correct matches and misses most particles, whereas the fine-tuned model recovers almost the complete ground-truth field with far fewer misses. (b) Quantitative comparison by validation tomogram. Four tomograms remain near $F_1 \approx 0.94$, while the noisiest volume, *tomo_8*, falls to $F_1 = 0.706$. The mean base performance remains near zero, whereas the fine-tuned model reaches mean $F_1 = 0.891$.

Interpretation and conclusions. The table in Figure 4.2(b) also exposes the main failure mode that remains after adaptation. Performance is strong and consistent in four tomograms, but *tomo_8* is visibly harder. Figure 4.2(a) shows the mechanism on a representative validation tomogram: fine-tuning changes the model from a detector that misses almost everything into one that covers the particle field with only a small number of residual FP and FN. After domain adaptation, the dominant issue is therefore no longer a failure to activate detections at all, but a loss of selectivity under difficult noise conditions. The conclusion is that fine-tuning is required under severe domain shift, and that post-adaptation robustness must be evaluated at the tomogram level rather than only through a global mean.

4.3. Most performance is recovered with few tomograms and multi-prompt inference reduces false positives

Question and hypothesis. Having established that fine-tuning is necessary under domain shift, the next question is how much annotated data is needed to recover most of the attainable performance, and whether prompt aggregation provides a measurable advantage over a single prompt. The hypothesis is that the learning curve will rise sharply at small sample sizes and then saturate, while multi-prompt inference will reduce FP by averaging away prompt-specific noise. The expected result is a low-data operating point and a multi-prompt advantage that appears mainly through precision.

Experiment summary. The experiment reuses exactly the same validation set as EXP2 and trains seven checkpoints with 1, 2, 4, 8, 12, 16, and 20 training tomograms. The adaptive-epoch policy ensures that small subsets receive longer optimization and are not penalized by construction.

The experiment also introduces the first fully explicit prompt-selection strategy of the thesis. The single-prompt branch uses a robust quality score based on three interpretable terms:

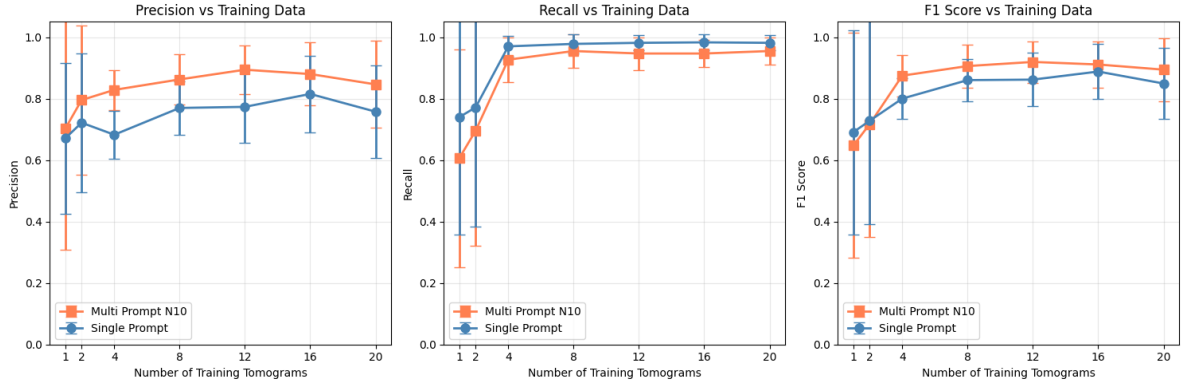
- A Fourier mid-band to high-band energy ratio, which favors structured signal over high-frequency noise.
- A center-to-border ratio, which favors crops whose main mass is centered.
- A DC penalty, which discourages crops dominated by offset or gradient artifacts.

The multi-prompt branch selects ten diverse crops by variance ranking, encodes them with TomoTwin and represents the class through both the first embedding and the mean embedding. The resulting prompt statistics show that the cosine similarity between the single and multi representations is 0.9629, with mean embedding-dimension variance 0.010789. This is important because it shows that the two prompt representations correspond to closely related instances of the same particle class; the expected gain is therefore robustness, not a change of class semantics.

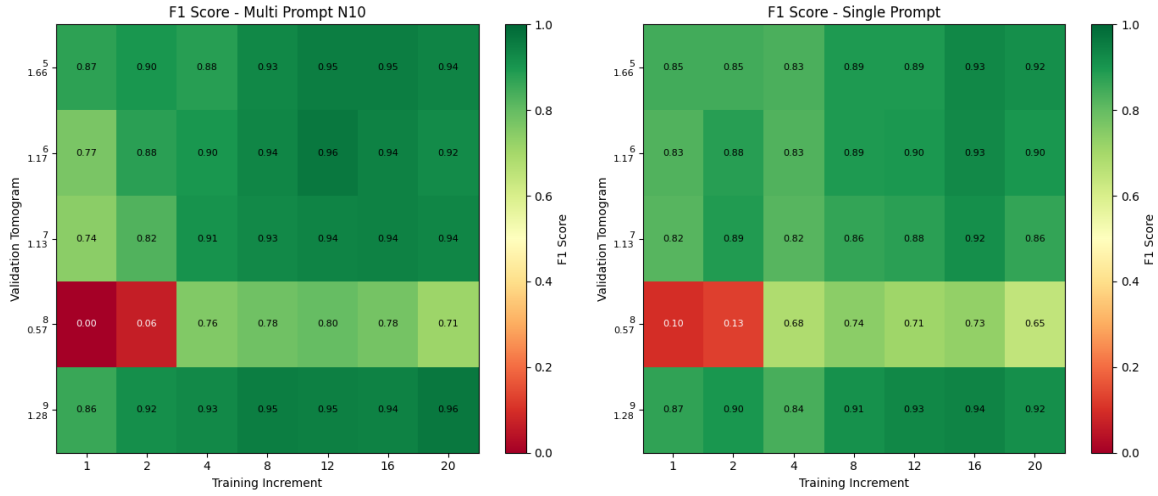
Results. Figure 4.3 now makes the EXP3 story visible at two complementary levels. Panel (a) shows the global learning curves: both strategies improve rapidly with few tomograms, the gain after 4 tomograms is already substantial, and the multi-prompt branch becomes clearly superior from that point onward. The best observed configuration is reached at 12 tomograms with multi-prompt inference, obtaining $F_1 = 0.919$, after which the curve largely saturates.

Panel (b) adds the tomogram-level view. Most validation tomograms stabilize at high F_1 once 4 to 8 training tomograms are available, but *tomo_8* remains consistently harder than the other four volumes. This matters because it shows that the aggregate curve is not hiding random noise: the learning trend is broadly shared across the validation set, while the residual weakness is concentrated in the same difficult tomogram. Panel (c) then explains why multi-prompt wins operationally. From 4 tomograms onward it usually gives up a small number of TP in exchange for a much larger FP reduction, which is exactly the tradeoff that improves precision and raises the final F_1 .

(a)



(b)



(c)

Training tomograms	Single prompt				Multi-prompt $n = 10$			
	F_1	TP	FP	FN	F_1	TP	FP	FN
1	0.691	468	153	154	0.647	384	60	238
2	0.728	488	124	134	0.717	440	54	182
4	0.800	604	285	18	0.875	578	118	44
8	0.860	609	186	13	0.906	595	95	27
12	0.862	611	190	11	0.919	590	70	32
16	0.888	612	149	10	0.911	590	83	32
20	0.849	611	220	11	0.894	595	118	27

Figure 4.3: EXP3 data efficiency and multi-prompt inference. (a) Mean precision, recall and F_1 as a function of training-set size. Most of the gain appears early and the multi-prompt branch overtakes the single-prompt branch from 4 tomograms onward, with the best observed result at 12 tomograms. (b) Validation-tomogram heatmaps for F_1 , showing that the improvement is broadly shared across the fixed split while *tomo_8* remains the persistent hard case for both prompt strategies. (c) Grouped summary with confusion totals extracted from the EXP3 inference notebook. The main multi-prompt advantage is not higher recall at all costs, but a strong FP reduction with only a moderate TP loss, which explains the better precision and final F_1 .

Interpretation and conclusions. If annotation time is limited, 4 tomograms already define a strong operating point. More importantly, the multi-prompt strategy reaches better results with less training data: with only 4 tomograms it already surpasses the single-prompt branch at 8 tomograms, and it stays ahead thereafter. If the priority is peak performance under the current protocol, 12 tomograms with multi-prompt inference are the best choice. Figure 4.3 shows that this recommendation is supported not only by the mean curve but also by the tomogram-level heatmap. The advantage is not an artifact of one unusually easy validation volume: all tomograms except *tomo_8* climb into a high-performance regime quickly, while *tomo_8* remains the systematic outlier under both prompt strategies.

The implication is important for Cryo-ET workflows. The reason why multi-prompt is preferable is not that it recovers more particles at all costs, but that it suppresses FP much more effectively. At the 12-tomogram setting, the single-prompt model yields 190 FP, while the multi-prompt model reduces them to 70, with only a moderate loss in TP. A detector with slightly lower recall but far fewer FP may be operationally preferable because it reduces manual curation and prevents contamination of downstream subtomogram processing. Methodologically, EXP3 also shows that prompt selection cannot remain an informal choice once the detector is already strong. The implementation of prompt ranking, prompt aggregation and tomogram-level failure analysis becomes part of the model-selection problem itself.

4.4. Prompt identity remains a failure mode after fine-tuning

Question and hypothesis. The last question is whether a fine-tuned multi-prompt checkpoint is invariant to the particular particles used as prompts. The hypothesis is that family-level averages can remain strong while a small number of prompts are poor representatives. In that case, the dominant failure should be recall collapse, not an increase in false detections, and no single scalar descriptor should fully identify the bad prompts. The expected result is a strong but imperfect detector whose remaining failures are prompt-specific.

Experiment summary. EXP4 has two parts. The first is the basic prompt-rotation experiment, which uses the 10 rotation-diverse prompts selected from training tomograms and evaluates how prompt identity changes performance for the *multi_inc16* checkpoint. The second is the larger prompt-population, where the thesis expands the analysis to 200 generated prompt candidates, using the first 125 for statistical inspection and rotation/acquisition diagnostics. In both parts, prompts are constructed only from training tomograms, avoiding validation leakage.

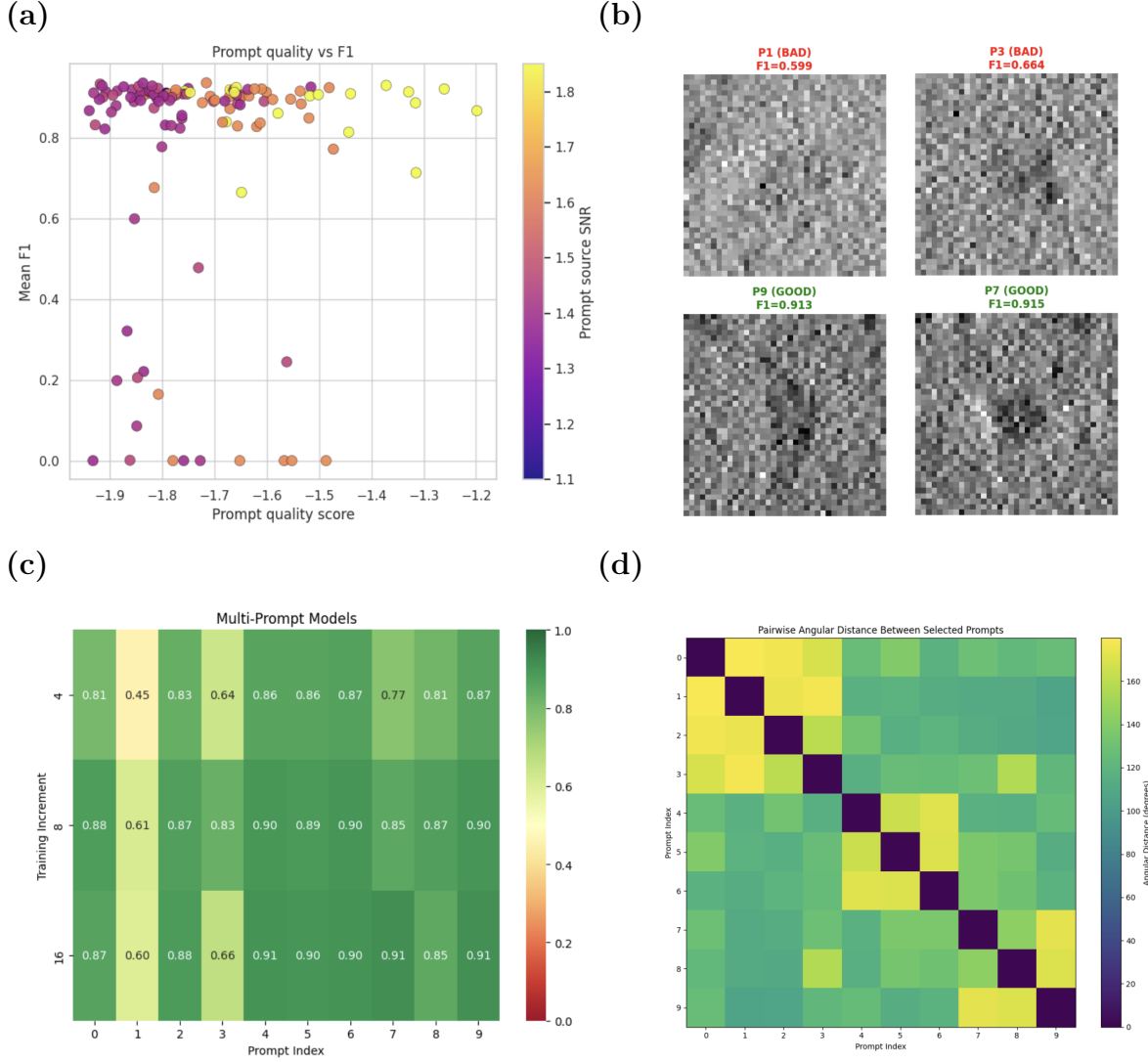
Prompt construction is explicit and staged:

1. Remove particles closer than 50 voxels to a tomogram border.
2. Compute the physical-quality score introduced in EXP3.
3. Keep the top-quality pool using a 65th-percentile threshold.
4. Select a rotation-diverse population by greedy quaternion distance.
5. Evaluate prompt-level behavior on the fixed five-tomogram validation set.

This construction first defines the controlled 10-prompt experiment and then scales the same logic to the larger population. Each candidate coordinate is checked so that a full 37^3 crop can be extracted, scored by frequency structure, centering and DC offset, and then selected for orientation diversity. Because thyroglobulin has C_2 symmetry, the distance is measured in $SO(3)/C_2$, so orientations related by the 180-degree symmetry operation are treated as equivalent. The practical effect is that EXP4 tests clean and plausible prompts that deliberately cover different views of the same particle class.

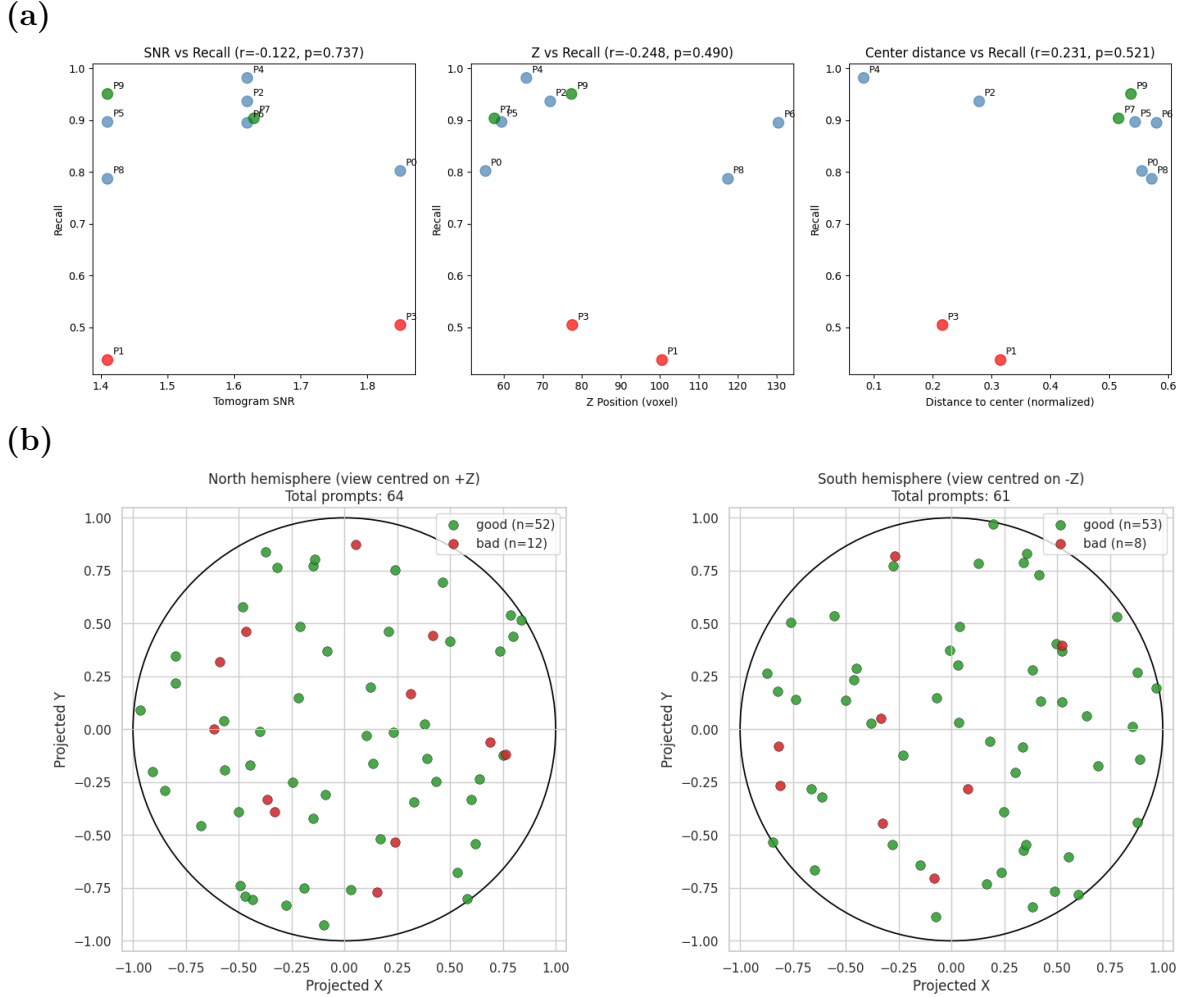
Results for basic prompt-rotation experiment. The first part of EXP4 asks whether the selected 10 prompts behave equivalently once the detector has already been fine-tuned. Figure 4.4 shows that they do not. Mean single-prompt performance remains high ($F_1 = 0.840 \pm 0.130$, precision = 0.912, recall = 0.809), but the average hides a clear spread: P7 and P9 reach $F_1 = 0.915/0.913$, while P1 and P3 fall to $F_1 = 0.599/0.664$. The hardest validation volume is again *tomo_8*, with mean $F_1 = 0.701$ and mean recall = 0.688.

The result indicates that the checkpoint is not globally weak; rather, a small number of plausible, rotation-diverse prompts are poor representatives for the validation distribution. Max aggregation gives high recall (0.9818) but lower precision (0.7067), producing $F_1 = 0.8169$. Removing P1 and P3 does not change the aggregate metric, suggesting that aggregation partly protects against isolated bad prompts, even though prompt identity still matters in the single-prompt setting.



Diagnostic	Result
Study design	10 rotation-diverse training prompts evaluated on the five validation tomograms with the <i>multi_inc16</i> checkpoint.
Single-prompt sweep	$F_1 = 0.840 \pm 0.130$, Precision = 0.912, Recall = 0.809.
Prompt spread	P1 and P3 are worst ($F_1 = 0.599/0.664$); P7 and P9 are best ($F_1 = 0.915/0.913$).
Hardest tomogram	<i>tomo_8</i> , mean $F_1 = 0.701$, mean Recall = 0.688.
Multi-prompt max	Max aggregation gives $F_1 = 0.8169$, Precision = 0.7067, Recall = 0.9818; removing P1 and P3 produces no metric change.

Figure 4.4: EXP4 basic prompt-rotation results for the *multi_inc16* checkpoint. (a) Prompt quality versus F_1 shows a broad high-performance plateau interrupted by sparse prompt collapses. (b) Central-slice inspection shows that weak prompts are not simply the noisiest crops, supporting a representativeness explanation. (c) Prompt-wise F_1 across training increments shows that most prompts stabilize at increments 8 and 16, but P1 and P3 stay weaker. (d) Pairwise angular distances confirm that the selected prompts cover a broad orientation range rather than ten near-duplicate particles.



Diagnostic	Result
Study design	200 prompt candidates generated from training tomograms only; first 125 prompts used for statistical inspection of <i>multi_inc16</i> .
Population performance	$F_1 = 0.780 \pm 0.283$, Precision = 0.824, Recall = 0.783; <i>tomo_8</i> is hardest, with mean $F_1 = 0.659$.
Acquisition descriptors	SNR, source Z and center distance have weak, non-significant correlations with recall ($ r \leq 0.248$, $p \geq 0.490$).
Orientation map	Bad prompts occur in both hemispheres: north has 64 prompts (52 good, 12 bad), south has 61 prompts (53 good, 8 bad).
Failure signature	Bad prompts have lower prediction load (0.4836 vs 1.0304), lower contrast (28.15 vs 92.95), lower center distance (0.2662 vs 0.5263) and larger mass-center shift (0.4179 vs 0.3258).
Follow-up candidates	The underperforming-prompt analysis finds 39 new candidates after excluding the original bad prompts; 10 are selected for a possible reproduction test.

Figure 4.5: EXP4 large prompt-population rotation and acquisition diagnostics. (a) Source SNR, source Z position and distance to tomogram center show weak, non-significant correlations with recall, so the bad prompts are not explained by any one scalar acquisition or position descriptor. (b) The local- Z orientation map places good and bad prompts in both hemispheres, ruling out a single global orientation rule.

Results for large prompt-population study. The second part asks whether the same prompt-specific failure mode remains visible when the analysis is expanded beyond the selected 10 prompts. Figure 4.5 summarizes the larger population study, using the first 125 prompts from the 200-prompt population generated from training tomograms only. The expanded sweep confirms that prompt variability is not an artifact of the small 10-prompt subset. Across the population, performance is much more dispersed, with $F_1 = 0.780 \pm 0.283$, precision = 0.824 and recall = 0.783.

The larger study also makes the failure signature more specific. Bad prompts do not appear to be explained by source SNR, source Z coordinate or distance to the tomogram center alone: all correlations with recall are weak and non-significant ($|r| \leq 0.248$, $p \geq 0.490$). The orientation map gives the same conclusion from a geometric perspective. Good and bad prompts appear in both hemispheres, with 52 good and 12 bad prompts in the north group and 53 good and 8 bad prompts in the south group. This rules out a simple explanation where one acquisition side or one global orientation class is responsible for the failures.

The remaining differences point instead to prompt representativeness and crop content. Bad prompts have lower prediction load, lower contrast, lower center distance and larger mass-center shift than good prompts. These descriptors suggest that weak prompts are not necessarily visually unusable, but their local structure is less effective as a conditioning example for the detector. The follow-up candidate search therefore identifies additional underperforming prompts for reproduction testing rather than treating the first weak prompts as isolated accidents.

Interpretation and conclusions. The basic prompt-rotation experiment shows that prompt identity still matters after fine-tuning. Figure 4.4(a) and the integrated summary table show a broad high-performance plateau interrupted by sparse prompt failures. These failures are mostly under-activation cases: the prompt does not produce enough correct detections on held-out tomograms, rather than producing a large number of FP.

The large prompt-population study makes the explanation narrower. Figure 4.5(a) shows that bad prompts are not explained by source SNR, Z coordinate or distance to the tomogram center alone. Figure 4.5(b) also rules out a simple «bad hemisphere» interpretation. This does not mean that rotation or acquisition geometry are irrelevant; it means that their effect is mixed with crop content, local reconstruction bias and the C2 symmetry of thyroglobulin [24].

The visual and angular analyses clarify the role of the multi-prompt input. Figure 4.4(b) shows that a bad prompt is not just an obviously noisy crop; relative to the good prompts, it also appears to have lower contrast. Figure 4.4(d) confirms that the selected prompts are rotation-diverse and Figure 4.4(c) shows that most selected prompts perform well once enough training data are used, while P1 and P3 remain weak. Therefore, the final TomoTwin input should not be understood as one perfectly representative particle. It is better understood as an averaged embedding of several clean and diverse particles, designed to dilute the risk that any single prompt is a poor representative of the class.

Chapter 5

Conclusions

5.1. Introduction

This thesis set out to evaluate whether a promptable DL detector can act as a practical step toward universal particle localization in Cryo-ET. The answer is positive, but conditional. ProPicker is useful when it is embedded in a disciplined workflow, especially when the target domain is fine-tuned and prompt construction is controlled. It is not yet universal in the stronger sense of transferring reliably to every new particle or acquisition condition from a single prompt and the base checkpoint alone.

5.2. Summary findings

The first finding is that prompt-only transfer is informative but incomplete. On EMPIAR-10988, the base model already detects ribosome signal, yet it does so with low recall. Threshold calibration improves the operating point only modestly, while fine-tuning produces the real change of regime and raises the validation F_1 to 0.5714. This result validates the repository pipeline and shows that the model contains useful transferable structure, but it also makes clear that the base model should not be treated as a final detector on real data without adaptation.

The second finding is that domain shift remains a hard limit. On the PolNet thyroglobulin dataset, the pretrained detector collapses to $F_1 \approx 0.008$, while the fine-tuned model reaches a mean F_1 of 0.891 on the fixed validation split. This is the clearest boundary found in the thesis: universality cannot be claimed from promptability alone. When the particle class, tomogram statistics or reconstruction conditions change substantially, fine-tuning is the step that makes the method operational.

The third finding is encouraging from an annotation-cost perspective. The incremental study shows that strong performance appears with relatively little training data. Four tomograms already define a useful operating point, and the best observed configuration is obtained with 12 training tomograms and multi-prompt inference, reaching $F_1 = 0.919$. The multi-prompt gain is also practical rather than cosmetic: it mainly reduces FP, which is exactly the type of improvement that makes later subtomogram extraction and manual inspection easier.

The fourth finding is that the main unresolved problem after fine-tuning is prompt robustness. EXP4 shows that a strong checkpoint can still fail for particular prompts, mostly through recall collapse rather than through excessive FP. The available diagnostics do not support a simple explanation based only on source SNR, Z position, distance

to tomogram center or a single orientation rule. The most defensible interpretation is that prompt quality depends on the joint effect of crop content, anisotropic acquisition, thyroglobulin symmetry and latent-space representativeness.

5.3. Limitations

These conclusions should be read within the scope of the experiments. The real-data component is a controlled proof of concept rather than a full EMPIAR-scale evaluation. The completed synthetic study is single-class, even though the repository is structured to support broader extensions. Some post-processing thresholds are inherited from the reference workflow and are kept fixed for comparability rather than rederived exhaustively in every setting. Finally, prompt selection is itself a source of variance, so the reported metrics must always be interpreted together with the prompt protocol used to obtain them.

5.4. Future work

The most immediate future direction is automatic prompt screening. A deployment-oriented workflow should rank candidates by structural quality, centering, diversity and representativeness before inference. A second direction is broader validation on additional real datasets, more particle families and multiclass settings. A third direction is to make adaptation cheaper through lightweight fine-tuning, active learning or human-in-the-loop refinement. A fourth direction is to incorporate Cryo-ET physics more directly, for example through missing-wedge-aware augmentation and symmetry-aware evaluation in spaces such as $SO(3)/C_2$.

The final conclusion is therefore deliberately qualified. ProPicker is not yet a universal Cryo-ET detector in the strict sense, but it is already a strong adaptable detector. With consistent preprocessing, limited fine-tuning, multi-prompt conditioning and explicit failure analysis, it provides a credible foundation for building more reliable universal detectors for macromolecule localization.

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Appendix A

Hardware Specifications and Software Stack

A.1. Documented experimental hardware

Cryo-ET fine-tuning and tomogram inference are computationally demanding tasks that depend on graphics processing unit (GPU) capacity, memory availability and storage throughput. For this reason, the experimental pipeline was executed on a dedicated workstation whose main specifications are summarized in Table A.1.

Component	Documented value
SSOO	Ubuntu 24.04.3 LTS (kernel 6.17.0-20-generic)
CUDA version	13.0
Main training GPU	NVIDIA GeForce RTX 4090 (24 GB VRAM)
CPU	13th Gen Intel(R) Core(TM) i7-13700KF
RAM	$\approx$ 135 GB

Table A.1: Experimental hardware information of the AIKE laboratory workstation.

A.2. Local authoring and validation environment

Part of the writing and repository inspection for this memory was carried out on the local system shown in Table A.2. This machine should not be confused with the documented experimental training workstation.

Component	Value
Computer	MacBook Pro (Mac14,10)
CPU	Apple M2 Pro, 12 CPU cores (8 performance + 4 efficiency)
Memory	16 GB unified memory
Integrated GPU	Apple M2 Pro, 19 GPU cores

Table A.2: Local authoring environment used for the present manuscript work.

A.3. Software, languages and technologies

The project relies on standard technologies widely used in scientific computing summarized in Table A.3, chosen for their mature documentation, compatibility with Cryo-ET data formats and straightforward integration with the ProPicker workflow.

Category	Tool or technology	Role in the project
Programming	Python	Main implementation language for preprocessing, fine-tuning, inference and analysis.
Manuscript	LaTeX	Preparation of the thesis manuscript.
Data formats	JSON, CSV, MRC	Prompt embeddings, coordinate tables and tomogram/localization-map storage.
Experiments	Jupyter Notebook	Exploratory preprocessing, metric analysis, prompt inspection and plotting.
Core detector	ProPicker	Promptable 3D segmentation model used as the central detector platform.
Training model	DeepETPicker	Generation of preprocessing and training configs and command-line training/inference execution.
Prompt encoder	TomoTwin	Extraction of prompt embeddings from $37 \times 37 \times 37$ subtomograms.
Main libraries	PyTorch, NumPy and Pandas	Checkpoint loading, inference-time handling of localization maps, numerical operations and table manipulation.
Volume input/output	<i>mrcfile</i> and toolchain MRC utilities	Reading and writing tomograms, labels and localization maps.
Visualization	ParaView	Inspection of tomograms and exported localization maps in volumetric form.
Environment	Conda environment	Execution of the ProPicker/DeepETPicker training and inference stack.
Version control	Git	Repository tracking.

Table A.3: Software, languages and technologies used.

Appendix B

Repository Structure and Implementation Details

The source code and implementation for this project are available in the public GitHub repository: `cryoet-particle-picking`.

B.1. Current repository scaffolding

The repository summarized in Table B.1 was designed for scalability to facilitate the seamless integration of future experiments.

Component	Main role	Practical guideline
<i>paths.py</i>	Central path registry	Update this file first when moving the project to a new machine. All experiment scripts depend on it.
<i>config.py</i>	Shared experimental constants	Keep common splits and hyperparameters here so that EXP2, EXP3 and EXP4 remain directly comparable.
<i>data/</i>	Input datasets	Store EMPIAR and PolNet tomograms, annotations and other raw inputs here.
<i>experiments/</i>	Experiment workspace	Store the specific experiment folders, notebooks and scripts here.
<i>results/</i>	Derived artifacts	Prompts, coordinates, checkpoints, inference outputs and analysis tables should be written here rather than into the source folders.
<i>docs/</i>	Documentation and reports	The thesis, interim reports and future experiment specifications are maintained here and versioned together with the code.
<i>models/</i>	Base checkpoints	Store all the checkpoints used for experiments.
<i>tools/</i>	Dependencies	Keep the cloned ProPicker here.

Table B.1: Repository scaffolding used by the thesis workflow.

B.2. Minimal reproducibility checklist

For a reader who wants to rerun or extend the investigation, the minimum checklist is the following:

1. Configure *paths.py* for the local machine.
2. Place the ProPicker and TomoTwin checkpoints in *models/*.
3. Clone ProPicker into *tools/ProPicker/*.
4. Install the Python environment used by DeepETPicker and ProPicker.
5. Populate *data/* with the EMPIAR or PolNet datasets.
6. Execute the notebook that generates prompts and coordinates for the target experiment.
7. Run the training or inference script corresponding to that experiment.
8. Verify scalar outputs and, when relevant, inspect the exported MRC maps in ParaView.

This checklist is intentionally minimal. Its role is to make the repository intelligible as a research instrument and to document the scaffolding assumptions on which the thesis results depend.

B.3. Condensed Algorithm Listings

The notebooks used in the experiments contain a large amount of supporting code for plotting, file management and interactive inspection. For the appendix, the most useful compromise is to keep only the core algorithmic fragments that define how prompts are scored, selected, aggregated and evaluated. The listings below are therefore condensed from the real experiment code rather than copied verbatim from entire notebooks. Their purpose is to preserve the research logic in a form that is short enough to audit.

B.3.1. Implementation coverage

Table B.2 links the main repository implementations to the condensed appendix listings. The training and inference wrappers are not reproduced in full because they mainly set paths, generate DeepETPicker configuration files and call ProPicker entry points. The algorithmic parts that affect the scientific conclusions are the prompt selectors, embedding aggregation and matching metrics, and those are included below.

Repository implementation	Appendix coverage	Reason for inclusion
<i>experiments/exp1_empiar10988_ribo/scripts/empiar10988_fine_tuning.py</i> and <i>empiar10988_inference.py</i>	Methodology text and EXP1 protocol	These scripts implement the real-data crop, contrast inversion, ProPicker training and inference wrapper.
<i>experiments/exp2_polnet_thy/scripts/polnet_fine_tuning.py</i> and <i>polnet_inference.py</i>	Methodology text and EXP2 protocol	These scripts implement the PolNet 20/5 split, voxel-space coordinates, contrast inversion and inference export.
<i>experiments/exp3_ppicker_limits/exp3_ppicker_limits.ipynb</i>	Listing B.1	This notebook contains the physical prompt-quality score used for single-prompt selection.
<i>experiments/exp3_ppicker_limits/exp3_ppicker_limits_inference.ipynb</i>	Listing B.2 and Listing B.4	This notebook builds single and multi-prompt embeddings, evaluates increments and computes metrics.
<i>experiments/exp4_ppicker_rotations/exp4_large_prompt_population_inference.ipynb</i>	Listing B.3 and Listing B.4	This notebook builds the large prompt population, applies quality filtering, uses C2-aware angular distances and evaluates prompt-level robustness.

Table B.2: Coverage of the important repository implementations in the appendix.

B.3.2. EXP3 prompt scoring and single-prompt selection

The core prompt-selection logic used in *experiments/exp3_ppicker_limits/exp3_ppicker_limits.ipynb* combines three physically motivated features: a mid-frequency versus high-frequency ratio, a center-versus-border energy ratio and a penalty for global offset. In the full notebook, helper functions perform robust normalization and optional embedding-consensus refinement; Listing B.1 keeps only the essential decision rule.

```

1 def compute_quality_score(subtomo, border=4, centre_frac=0.45,
2                           k_low=0.08, k_high=0.30):
3     vol = robust_norm(to_numpy(subtomo))
4     freq_ratio = freq_structure_ratio(vol, k_low=k_low, k_high=k_high)
5     centre_ratio = center_border_ratio(
6         vol, border=border, centre_frac=centre_frac
7     )
8     dc_penalty = abs(vol.mean()) / (vol.std() + 1e-6)
9
10    score = np.log(freq_ratio) + 0.9 * np.log(centre_ratio) - 0.6 * dc_penalty
11    return score, {
12        "freq_ratio": freq_ratio,
13        "centre_ratio": centre_ratio,
14        "dc_penalty": dc_penalty,
15    }
16
17
18 def select_best_prompt(subtomos, prompt_encoder=None, top_k=8):
19     base_scores = np.array([compute_quality_score(v)[0] for v in subtomos])

```

```

20 candidate_idx = np.argsort(base_scores)[::-1][:top_k]
21 final_scores = base_scores.copy()
22
23 if prompt_encoder is not None and len(candidate_idx) >= 2:
24     E = np.stack([prompt_encoder(subtomos[i]).reshape(-1)
25                  for i in candidate_idx])
26     S = cosine_similarity_matrix(E)
27     consensus = (S.sum(axis=1) - 1.0) / (S.shape[0] - 1)
28     for j, i in enumerate(candidate_idx):
29         final_scores[i] += 1.2 * consensus[j]
30
31 best_idx = int(np.argmax(final_scores))
32 return best_idx, subtomos[best_idx]

```

Listing B.1: Condensed EXP3 single-prompt selector based on subtomogram quality.

This selector reflects an important methodological shift in the thesis: once detector performance becomes competitive, prompt choice itself becomes part of model selection rather than a manual convenience step.

B.3.3. EXP3 multi-prompt aggregation

The second important idea introduced in *experiments/exp3_ppicker_limits/exp3_ppicker_limits_inference.ipynb* is that the operational prompt does not have to be a single subtomogram embedding. In the experiment, several prompt embeddings are computed from selected particles and their arithmetic mean is used as the «multi-prompt» representation. Listing B.2 shows the compact form of that logic.

```

1 def build_prompt_embeddings(subtomos, prompt_encoder, n_instances=10):
2     embeddings = torch.stack([
3         prompt_encoder(subtomo).reshape(-1)
4         for subtomo in subtomos[:n_instances]
5     ])
6
7     single_prompt = embeddings[0]
8     multi_prompt = embeddings.mean(dim=0)
9
10    return {
11        "single_prompt": single_prompt,
12        "multi_prompt": multi_prompt,
13        "all_embeddings": embeddings,
14    }

```

Listing B.2: Condensed EXP3 multi-prompt aggregation. In the notebook, the input subtomograms are already pre-selected by the prompt-scoring stage.

In the actual experiment, this averaged embedding produced the best observed operating mode because it reduced FP more strongly than it reduced recall.

B.3.4. EXP4 Step 2: quality-filtered rotation-diverse prompt selection

Step 2 of *experiments/exp4_ppicker_rotations/exp4_large_prompt_population_inference.ipynb* extends prompt selection by adding quaternion distances and a

symmetry-aware interpretation for thyroglobulin. The selector first filters out unsafe border candidates, then keeps only high-quality subtomograms, and finally builds a prompt population that is as diverse as possible in orientation space while still preferring better-quality candidates. Listing B.3 gives the condensed form of the notebook logic.

```

1 def quaternion_angular_distance_c2(q1, q2, c2_rotation):
2     r1 = Rotation.from_quat(normalize_quaternion(q1))
3     r2 = Rotation.from_quat(normalize_quaternion(q2))
4     return min(
5         np.degrees((r1.inv() * r2).magnitude()),
6         np.degrees((r1.inv() * (r2 * c2_rotation)).magnitude()),
7     )
8
9
10 def select_diverse_rotations_with_quality(
11     df, tomo_cache, n_samples, min_angular_dist=30.0,
12     quality_percentile=65.0, max_per_tomo=None
13 ):
14     df = filter_border_safe_candidates(df, tomo_cache, min_border_margin=50)
15     df["quality_score"] = [
16         compute_quality_score(extract_subtomo(row, tomo_cache))[0]
17         for _, row in df.iterrows()
18     ]
19
20     threshold = np.nanpercentile(df["quality_score"], quality_percentile)
21     high_quality = df[df["quality_score"] >= threshold].copy()
22     if len(high_quality) < n_samples:
23         threshold = np.sort(df["quality_score"].to_numpy())[-n_samples]
24         high_quality = df[df["quality_score"] >= threshold].copy()
25     df = high_quality
26
27     quats = df[["Q1", "Q2", "Q3", "Q4"]].to_numpy()
28     scores = df["quality_score"].to_numpy()
29     tomo_names = df["tomo_name"].to_numpy()
30
31     selected = [int(np.argmax(scores))]
32     tomo_counts = {tomo_names[selected[0]]: 1}
33
34     while len(selected) < min(n_samples, len(df)):
35         best_idx, best_min_dist, best_quality = -1, -1.0, -np.inf
36         for i in range(len(df)):
37             if i in selected:
38                 continue
39             if (
40                 max_per_tomo is not None
41                 and tomo_counts.get(tomo_names[i], 0) >= max_per_tomo
42             ):
43                 continue
44
45             min_dist = min(
46                 quaternion_angular_distance_c2(
47                     quats[i], quats[j], THYROGLOBULIN_C2_ROTATION
48                 )
49                 for j in selected
50             )
51
52             if min_dist >= min_angular_dist and (

```

```

53         min_dist > best_min_dist or
54         (min_dist == best_min_dist and scores[i] > best_quality)
55     ):
56         best_idx, best_min_dist, best_quality = i, min_dist, scores[i]
57
58     if best_idx == -1:
59         for i in range(len(df)):
60             if i in selected:
61                 continue
62             if (
63                 max_per_tomo is not None
64                 and tomo_counts.get(tomo_names[i], 0) >= max_per_tomo
65             ):
66                 continue
67             min_dist = min(
68                 quaternion_angular_distance_c2(
69                     quats[i], quats[j], THYROGLOBULIN_C2_ROTATION
70                 )
71             for j in selected
72             )
73             if min_dist > best_min_dist or (
74                 min_dist == best_min_dist and scores[i] > best_quality
75             ):
76                 best_idx, best_min_dist, best_quality = i, min_dist, scores[i]
77         if best_idx == -1:
78             break
79
80     selected.append(best_idx)
81     tomo_counts[tomo_names[best_idx]] = (
82         tomo_counts.get(tomo_names[best_idx], 0) + 1
83     )
84
85     return df.iloc[selected].copy()

```

Listing B.3: Condensed EXP4 selector for quality-filtered and rotation-diverse prompts.

The key research idea here is not only the use of quaternions, but the fact that the selector is symmetry-aware and quality-aware at the same time. This is precisely what made it possible to separate prompt quality effects from naive «bad rotation» explanations in the final analysis.

B.3.5. Fixed greedy evaluation rule used in the synthetic experiments

Both *exp3_ppicker_limits_inference.ipynb* and *exp4_large_prompt_population_inference.ipynb* keep a fixed center-matching rule between predicted coordinates and ground truth inside each analysis. EXP3 uses 18.5 voxels, equal to half the 37³ prompt size. EXP4 uses 15.0 voxels in the large prompt-population analysis to make prompt-level failures stricter. Listing B.4 shows the compact implementation. The matching is greedy and one-to-one: once a ground truth point has been matched, it cannot be reused by another prediction.

```

1 def compute_metrics(pred_coords, gt_coords, distance_threshold):
2     if len(pred_coords) == 0:
3         return {"precision": 0.0, "recall": 0.0, "f1": 0.0,

```

```

4         "tp": 0, "fp": 0, "fn": len(gt_coords)}
5     if len(gt_coords) == 0:
6         return {"precision": 0.0, "recall": 0.0, "f1": 0.0,
7               "tp": 0, "fp": len(pred_coords), "fn": 0}
8
9     distances = cdist(pred_coords, gt_coords)
10    matched_gt = set()
11    tp = 0
12
13    for i in range(len(pred_coords)):
14        best_j, best_d = -1, np.inf
15        for j in range(len(gt_coords)):
16            if j not in matched_gt and distances[i, j] < best_d:
17                best_j, best_d = j, distances[i, j]
18
19        if best_j != -1 and best_d <= distance_threshold:
20            matched_gt.add(best_j)
21            tp += 1
22
23    fp = len(pred_coords) - tp
24    fn = len(gt_coords) - tp
25    precision = tp / (tp + fp) if tp + fp else 0.0
26    recall = tp / (tp + fn) if tp + fn else 0.0
27    f1 = (2 * precision * recall / (precision + recall)
28         if precision + recall else 0.0)
29
30    return {"precision": precision, "recall": recall, "f1": f1,
31          "tp": tp, "fp": fp, "fn": fn}

```

Listing B.4: Condensed greedy one-to-one coordinate matching used to compute precision, recall and F_1 .

Preserving this rule across the synthetic experiments matters scientifically: it keeps the dataset constant while the prompt strategy, checkpoint family, or number of training tomograms changes.